Safe USV Recovery Using Disturbance-Aware Predictive Control with Control Barrier Functions

Kiyong Park, Changyu Lee, Dong-Hoon Kim, Woong-Ki Lee, Ji-Wan Kim, Jun-Hyuk Choi, and Jinwhan Kim

***Abstract*—This paper presents a safe autonomous recovery framework that integrates disturbance observer-based model predictive control (DOB-MPC) and robust adaptive control barrier functions (RaCBFs) for docking an underactuated unmanned surface vehicle (USV) into a side-mounted cradle on a mothership. Autonomous recovery is challenging because the underactuated USV, driven by a single stern thruster and rudder, must thread a narrow cradle entrance without colliding with the mothership while rejecting persistent wind, wave, and current disturbances. To achieve robust and dynamically feasible tracking, the DOB-MPC embeds a disturbance estimate into the MPC prediction model while respecting the underactuated dynamics and actuator limits. For safe recovery, the operation is divided into an approach and dynamic-positioning phase and a docking phase, and phase-specific RaCBFs, a side-boundary barrier and a funnel-shaped docking barrier, are tightened online by the DOB error bound and embedded across the MPC prediction horizon so that safety is enforced over the entire horizon rather than one step ahead. The framework is validated through Monte Carlo simulations and real-world experiments under scaled wave and current disturbances, where it achieves the highest recovery success rate and the fewest safety-constraint violations among the compared baseline controllers.**

***Index Terms*—Autonomous surface vehicle, control barrier function, disturbance observer, launch and recovery system, model predictive control.**

I. Introduction

Maritime operations are often performed using collaborative systems composed of a large mothership and small auxiliary vessels. Such systems are widely utilized in missions including surveillance, search and rescue, environmental monitoring, transportation of supplies, and offshore operations, where the mothership provides transportation and operational support while smaller vessels perform mission-specific tasks. These systems commonly adopt a marsupial architecture, in which the mothership deploys and recovers small vessels during operation through a launch and recovery system (LARS). To effectively utilize unmanned surface vessels (USVs) within such marsupial systems, autonomous launch and recovery capability is essential. Among various autonomous recovery configurations, cradle-based recovery systems mounted on the side of the mothership are widely adopted due to their structural simplicity and operational practicality.

However, autonomous recovery of USVs into a cradle remains a highly challenging problem due to several coupled factors. First, USVs operate under persistent environmental disturbances such as wind, waves, and currents, which significantly degrade tracking accuracy during recovery maneuvers. Second, most small USVs are underactuated and subject to actuator saturation limits, making precise lateral positioning difficult during maneuvering. Third, the narrow cradle entrance requires accurate position and heading alignment while simultaneously avoiding collision with the mothership and cradle structure. Therefore, safe autonomous recovery requires both disturbance-aware trajectory tracking and online safety enforcement under constrained underactuated dynamics.

Model predictive control (MPC) is well suited to this setting, as it handles the vessel dynamics, input constraints, and underactuation while optimizing future actions [1]–[3], but as a model-based method its accuracy degrades under the environmental disturbance. Integrating a disturbance observer (DOB) into the MPC compensates the disturbance online [4], [5]. Because the DOB estimate is imperfect, we further establish a closed-loop boundedness guarantee that explicitly accounts for the DOB error.

While robust trajectory tracking is essential, tracking performance alone is insufficient to guarantee safe recovery operation. During recovery, the USV must avoid collision with the mothership and cradle boundaries while threading the narrow docking corridor, which calls for a safety mechanism that is both robust to disturbances and flexible enough to plan the ingress through the confined corridor.

Robust tube MPC, one way to enforce safety under disturbance, bounds the deviation induced by disturbance with a robust invariant set, but for the nonlinear underactuated vessel this set must be constructed offline from an incremental Lyapunov function or contraction metric over the maneuvering envelope [6]–[8]. Control barrier functions (CBFs) instead enforce safety as forward invariance through a barrier condition needing no such offline construction [9], [10]. A nominal CBF presumes undisturbed dynamics, so its invariance no longer holds under the persistent wind, wave, and current of recovery, and we instead adopt robust adaptive control barrier functions (RaCBFs) [11], [12] that tighten the barrier condition online with the DOB estimate to restore forward invariance under the disturbance, embedded across the MPC prediction horizon to plan the constrained maneuver.

Motivated by these challenges, this paper proposes a safe autonomous recovery framework that integrates DOB-MPC and phase-specific RaCBFs for underactuated USVs

This research was supported by. *(Corresponding author: Jinwhan Kim)*

Kiyong Park and Jinwhan Kim are with the Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34141, Republic of Korea (e-mail: qkrrldyd777@kaist.ac.kr; jinwhan@kaist.ac.kr).

Changyu Lee is with the Department of Mechanical and Automotive Engineering, Kongju National University, Cheonan 31080, Republic of Korea (e-mail: leeck@kongju.ac.kr).

Dong-Hoon Kim, Woong-Ki Lee, Ji-Wan Kim, and Jun-Hyuk Choi are with LIGDnA, Republic of Korea (e-mail: donghoon.kim@ligdna.com; woongki.lee@ligdna.com; jiwan.kim@ligdna.com; junhyuk.choi@ligdna.com).

operating under environmental disturbances. The proposed framework divides the recovery process into two phases: an approach/dynamic-positioning (DP) phase and a docking phase. During the first phase, the USV safely approaches the side of the mothership and maintains a desired relative position for stabilization. Once the relative pose becomes sufficiently stabilized, the controller transitions to the docking phase, where the USV enters the cradle while satisfying docking safety constraints. To support these objectives, phase-specific reference tracking points and RaCBFs are designed according to the operational characteristics of each recovery phase. The effectiveness of the proposed framework is validated through numerical simulations and pool experiments under scaled environmental disturbances.

The main contributions of this paper are summarized as follows:

- A DOB-MPC framework is proposed for underactuated USV recovery, with a closed-loop state boundedness result under bounded disturbance estimation error.
- A funnel-shaped docking CBF and a side-boundary CBF are formulated as phase-specific RaCBFs to ensure safe recovery under disturbances.
- The proposed recovery framework is validated through numerical simulations and real-world experiments under environmental disturbances.

The remainder of this paper is organized as follows. Section II reviews related works. Section III presents the system modeling of the USV. Section IV describes the proposed control framework including DOB-MPC, RaCBFs and recovery scenario. Section V presents numerical simulation results, followed by real-world experimental validation in Section VI. Finally, conclusions are provided in Section VII.

II. Related Work

Autonomous launch and recovery of marine vehicles from a mothership has been studied across various system configurations and control approaches. Most existing studies focus on AUV recovery from a USV platform, including waypoint-based guidance validated through sea trials but susceptible to wave and current disturbances [13], layered guidance with model-free adaptive control for handling external interference without safety constraints [14], and observer-based disturbance compensation combined with backstepping sliding mode control for mobile docking without collision-avoidance enforcement [15]. In contrast, recovery of a USV into a moving mothership has been comparatively less researched, with existing work primarily addressing kinematic path planning without accounting for environmental disturbances or safety constraints [16]. These results indicate that disturbance rejection and safety constraint enforcement have not been simultaneously addressed in the context of autonomous USV recovery operations.

MPC has been widely adopted for trajectory tracking of marine surface vehicles due to its ability to explicitly handle system dynamics and input constraints [1]–[3], [17], [18]. However, as a model-based strategy, its performance degrades under environmental disturbances and model uncertainties commonly encountered in maritime environments. To improve robustness, DOB-based approaches have been studied across various platforms, with two primary integration strategies: direct input compensation as in L1-adaptive control [17], [19], [20], and embedding DOB estimates into the controller dynamics [4], [5]. For marine vehicles, the former is not well suited due to unmatched uncertainties from underactuated dynamics and slow actuator response, while the latter has been integrated within MPC to compute disturbance-aware control inputs under explicit constraints. However, existing DOB-MPC formulations employ continuous-time disturbance observers without explicit discrete-time formulation, which limits reliable disturbance estimation at the relatively low sensor update rates characteristic of USV operations. In contrast, the proposed DOB-MPC employs a piecewise-constant adaptation law [19] with an explicit discrete-time formulation that incorporates the sampling interval, and a closed-loop state boundedness result is established to provide a theoretical robustness guarantee for safety-critical recovery operations.

Beyond robust tracking, safe recovery requires the USV to avoid collision with the mothership while threading the narrow cradle corridor. Tube-based MPC enforces safety against bounded disturbances by tightening the constraints with a robust invariant set that bounds the disturbance-induced deviation [6]–[8], [21]–[23], but this invariant set, for the nonlinear underactuated vessel, must be constructed offline from an incremental Lyapunov function or contraction metric over the maneuvering envelope. CBFs instead enforce safety as forward invariance of the safe set through a per-step condition on the barrier. The condition renders the set forward invariant for all subsequent time while requiring no offline invariant-set construction. CBFs have been widely adopted for safe docking and landing maneuvers [24]–[26]. These CBF formulations, however, assume disturbance-free dynamics or accommodate disturbances only implicitly, so their forward-invariance guarantee no longer holds once persistent disturbances act on the vehicle. RaCBFs restore safety by tightening the barrier condition with an uncertainty-dependent margin [11], [12], [27], but they are predominantly formulated for parametric model uncertainty and enforced as a one-step quadratic-program safety filter. Reacting one step at a time, such a filter presumes an actuator-admissible safe input always exists at the current instant, yet the vessel's actuator saturation and underactuation can empty this safe input set before the cradle is reached [28], [29]. Embedding CBF constraints within the MPC optimization instead propagates safety across the entire prediction horizon [30], [31], yet such MPC-CBF formulations assume disturbance-free dynamics and offer no robustness to environmental disturbance. The proposed framework brings these strengths together. Phase-specific RaCBFs, whose disturbance margins are supplied online by the DOB, are embedded across the MPC prediction horizon, combining horizon-wide safety planning with robustness to the environmental disturbance.

III. Dynamic Modeling of Surface Vehicles

The motion of the USV is described by maneuvering model of [32], consisting of a kinematic equation that relates the

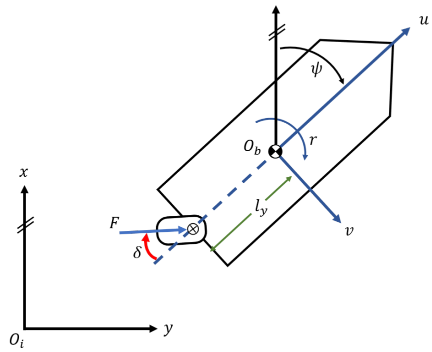

Fig. 1. Coordinate of the USV.

body-fixed and world-fixed frames and a kinetic equation that governs the body-fixed velocity. The kinematics express the world-frame pose $\eta = [x, y, \psi]^\top$ in terms of the body-fixed velocity $\nu = [u, v, r]^\top$ through the yaw rotation matrix $R(\psi)$.

$$\dot{\eta} = R(\psi)\,\nu, \qquad R(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{1}$$

The body-fixed kinetics balance the inertial, Coriolis, damping, control, and disturbance terms:

$$M\,\dot{\nu} + C(\nu)\,\nu + D\,\nu = \tau_c + \tau_w, \tag{2}$$

where M is the inertia matrix, $C(\nu)$ the Coriolis–centripetal matrix, D the hydrodynamic damping matrix, τ_c the control force and moment, and $\tau_w = [\tau_{w,u}, \tau_{w,v}, \tau_{w,r}]^\top$ the external disturbance force and moment. These matrices are given by

$$M = \mathrm{diag}(m_{11},\, m_{22},\, m_{33}), \tag{3a}$$

$$C(\nu) = \begin{bmatrix} 0 & 0 & -m_{22}v \\ 0 & 0 & m_{11}u \\ m_{22}v & -m_{11}u & 0 \end{bmatrix}, \tag{3b}$$

$$D = -\begin{bmatrix} X_u & 0 & 0 \\ 0 & Y_v & Y_r \\ 0 & N_v & N_r \end{bmatrix} - \begin{bmatrix} X_{u|u|}|u| & 0 & 0 \\ 0 & Y_{v|v|}|v| & Y_{r|r|}|r| \\ 0 & N_{v|v|}|v| & N_{r|r|}|r| \end{bmatrix}, \tag{3c}$$

in which $m_{11} = m - X_{\dot{u}}$, $m_{22} = m - Y_{\dot{v}}$, and $m_{33} = I_z - N_{\dot{r}}$ are the effective masses and yaw moment of inertia with the hydrodynamic added-mass terms included, the first matrix in (3c) collects the linear drag coefficients, and the second collects the quadratic drag coefficients.

The USV is actuated by a single stern outboard engine as shown in Fig 1, whose control force and moment $\tau_c = [\tau_u, \tau_v, \tau_r]^\top$ are set by the throttle input n_T, and the steering input n_S, which sets the outboard-engine deflection angle:

$$\begin{bmatrix} \tau_u \\ \tau_v \\ \tau_r \end{bmatrix} = \begin{bmatrix} c\,n_T^2 \cos(\alpha n_S) \\ c\,n_T^2 \sin(\alpha n_S) \\ -\,l_y\,c\,n_T^2 \sin(\alpha n_S) \end{bmatrix}, \tag{4}$$

where c is the thrust control coefficient, α is the steering-input-to-engine-angle coefficient, and l_y is the distance from the origin of the body-fixed frame to the outboard engine.

The throttle and steering inputs cannot be commanded arbitrarily. Their amplitudes are confined to the admissible set

$$\mathcal{U}_c = \left\{ \mathbf{n} = [n_S, n_T]^\top \in \mathbb{R}^2 \;\middle|\; \mathbf{n}_{\min} \le \mathbf{n} \le \mathbf{n}_{\max} \right\}, \tag{5}$$

with the inequality understood entrywise, and their finite actuation speed additionally limits the command rates:

$$\dot{\mathcal{U}}_c = \left\{ \dot{\mathbf{n}} = [\dot{n}_S, \dot{n}_T]^\top \in \mathbb{R}^2 \;\middle|\; \dot{\mathbf{n}}_{\min} \le \dot{\mathbf{n}} \le \dot{\mathbf{n}}_{\max} \right\}. \tag{6}$$

IV. Control System Design

The proposed framework, shown in Fig. 2, executes the cradle recovery in two sequential phases. Each phase activates a different set of phase-specific RaCBFs and a different MPC reference, coordinated by a two-phase recovery logic. The architecture is composed of this two-phase logic and three control components: a DOB that estimates the environmental disturbance via a piecewise-constant adaptation law, a DOB-MPC that tracks the phase-dependent reference by incorporating the estimated disturbance into its prediction model under the actuator amplitude and rate limits, and a set of RaCBFs that enforce safety as MPC constraints. The remainder of this section first introduces the two-phase recovery logic, then details the three components in turn.

A. Two-phase recovery

Cradle recovery imposes two safety-critical maneuvering objectives of different character: a transverse approach to the mothership side that must avoid collision with the mothership hull, and a longitudinal ingress into the cradle that must keep the vehicle within the narrow cradle corridor. Because the USV is underactuated, it cannot dedicate enough independent control authority to enforce both objectives at once. Attempting the recovery in a single phase therefore forces the limited authority into a trade-off between the two safety objectives. The recovery is consequently executed in two phases, each concentrating the control authority on one objective, an approach/DP phase and a docking phase. This two-phase logic is realized by the flowchart in Fig. 3.

Phase 1: approach and DP. The USV first approaches the mothership and enters the DP region. The MPC reference is the DP point at a fixed offset from the cradle in the mothership frame, and the active safety constraint is the side barrier CBF Φ_{side}. Once the USV holds the DP point for a continuous dwell time exceeding a threshold N_{dwell}, the controller transitions to Phase 2.

Phase 2: docking. The USV initiates the cradle recovery. The MPC reference switches to the cradle center, and the funnel docking barrier CBFs $\Phi_{\mathrm{dock,L}}, \Phi_{\mathrm{dock,R}}$ are activated alongside Φ_{side}. If the head point enters the cradle, the recovery completes. The USV may occasionally exit the safe region, either because the in-horizon RaCBF does not always guarantee recursive feasibility of the constrained optimization, or because of transient signal dropouts and unmodeled disturbance excursions. In that case, control returns to Phase 1 at the DP point to re-stabilize before retrying, which serves as a practical safeguard.

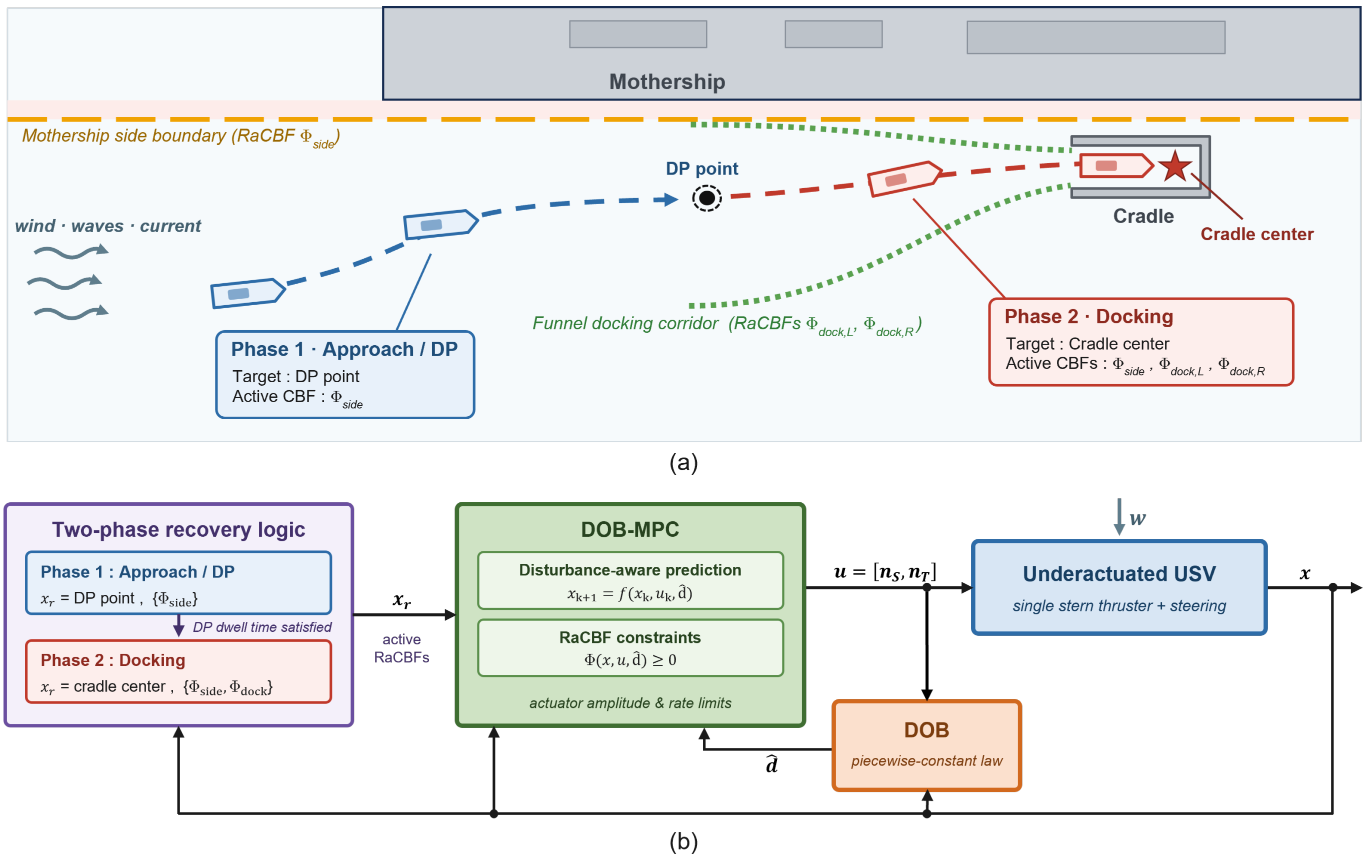

Fig. 2. (a) Two-phase recovery scenario: Phase 1 approaches the mothership side and holds at the DP point under the side barrier Φ_{side}. Phase 2 enters the cradle through the funnel docking corridor under the docking barriers $\Phi_{\mathrm{dock,L}}, \Phi_{\mathrm{dock,R}}$. (b) Safety-aware control architecture: a two-phase logic switches the active RaCBFs and the MPC reference between Phase 1 and Phase 2. DOB-MPC tracks the phase-dependent reference using a disturbance-aware prediction with RaCBF constraints.

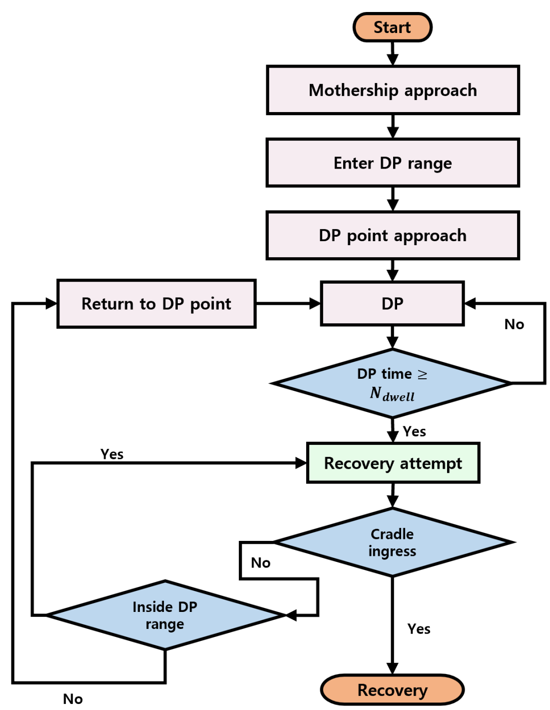

Fig. 3. Two-phase recovery state machine. Phase 1 drives the USV to the DP point and holds station until the dwell-time threshold is met, after which Phase 2 attempts cradle ingress with safe return to the DP point on failure.

B. DOB

To estimate the body-frame disturbance τ_w acting on the USV, we employ a piecewise-constant adaptation law as the DOB, which originates from the $\mathcal{L}_1$ adaptive control scheme [19].

The DOB operates on the surge, sway, and yaw velocities $\nu = [u, v, r]^\top$. From the kinetics (2), the velocity sub-system driven by the disturbance is written as

$$\dot{\nu} = f(\nu, \mathbf{u}) + \mathbf{w}, \tag{7}$$

where $f(\nu, \mathbf{u})$ denotes the nominal dynamics and the acceleration-level disturbance is

$$\begin{aligned} \mathbf{w} = M^{-1}\tau_w &= \left[\, \tau_{w,u}/m_{11},\ \tau_{w,v}/m_{22},\ \tau_{w,r}/m_{33} \,\right]^\top \\ &= \left[\, w_u,\ w_v,\ w_r \,\right]^\top. \end{aligned} \tag{8}$$

A state predictor is run in parallel with the true system:

$$\dot{\hat{\nu}} = f(\nu, \mathbf{u}) + \hat{\mathbf{w}} + A_e\, \tilde{\nu}, \tag{9}$$

where $\hat{\nu}$ is the predicted state, $\nu = \hat{\nu} - \nu$ is the prediction error, $\hat{\mathbf{w}}$ is the current disturbance estimate, and A_e is a Hurwitz adaptation gain. Discretizing (9) with sampling period T_s, the discrete-time state predictor is formulated as

$$\hat{\nu}_{k+1} = \hat{\nu}_k + \left[\, f(\nu_k, \mathbf{u}_k) + \hat{\mathbf{w}}_k + A_e\, \tilde{\nu}_k \,\right] T_s, \tag{10}$$

where k is the discrete time index and T_s is the DOB sampling period.

The piecewise-constant adaptation law updates the estimate at each sampling instant so that the prediction error is driven to zero over one interval. The estimate is expressed as

$$\hat{\mathbf{w}}_k = -A_e^{-1}\big(e^{A_e T_s} - I\big)^{-1} e^{A_e T_s}\,\tilde{\nu}_k. \tag{11}$$

Using the estimated disturbance, the final estimate $\hat{d}$ fed to the controller is obtained by applying a discretized first-order low-pass filter:

$$\hat{d}_k = e^{-\omega_c T_s}\,\hat{d}_{k-1} + \big(1 - e^{-\omega_c T_s}\big)\,\hat{\mathbf{w}}_k, \tag{12}$$

where ω_c is the cutoff frequency. Filtering is essential for safe operation, since the rapidly adapting law produces a raw estimate $\hat{\mathbf{w}}_k$ whose high-frequency content would otherwise make the control input and the RaCBF safety margin chatter. The low-pass filter confines the estimate to the low-frequency band on which the controller can act safely, so that fast adaptation improves robustness without itself becoming a source of unsafe behavior. The filtered estimate $\hat{d}$ is the single quantity fed into both the MPC prediction model and the RaCBF constraint, and its estimation-error bound is derived next.

1) Estimation error bound: The DOB estimate $\hat{d}$ contains an estimation error relative to the true disturbance $\mathbf{w}$. This subsection establishes the bound μ_w on this error, which subsequently sizes the residual set of the closed-loop robustness result (Section IV-C1) and the RaCBF safety margin (Section IV-D). The bound relies on two mild regularity assumptions on the disturbance and the nominal dynamics.

Assumption 1 (Bounded disturbance)**.** *The disturbance and its variations are bounded as*

$$\|\mathbf{w}\| \le \Delta_w, \quad \|\partial\mathbf{w}/\partial t\| \le L_{w_t}, \quad \|\partial\mathbf{w}/\partial\nu\| \le L_{w_\nu}.$$

Assumption 2 (Bounded nominal dynamics)**.** *The nominal dynamics are bounded as*

$$\|f(\nu,\mathbf{u})\| \le \Delta_f.$$

Under Assumptions 1–2, [19, Proposition 4] shows that the raw estimate $\hat{\mathbf{w}}$ produced by the adaptation law (11) satisfies

$$\|\mathbf{w} - \hat{\mathbf{w}}\| \le \begin{cases} \Delta_w, & 0 \le t < T_s, \\ \zeta\,T_s, & t \ge T_s, \end{cases} \tag{13}$$

with

$$\phi := \Delta_f + \Delta_w, \tag{14}$$

$$\zeta := 2\sqrt{3}\,(\phi\,L_{w_\nu} + L_{w_t}) + \sqrt{3}\,|A_e|\,\Delta_w, \tag{15}$$

specialized from [19, Eq. (30b),(30d)] to the three-dimensional velocity sub-system (7).

The MPC and the RaCBF, however, consume the filtered estimate $\hat{d} = C(s)\hat{\mathbf{w}}$ with $C(s) = \omega_c/(s+\omega_c)$. Let

$$\tilde{\mathbf{w}} := \mathbf{w} - \hat{d}, \tag{16}$$

denote the controller-facing estimation error. Adding and subtracting $C(s)\mathbf{w}$,

$$\tilde{\mathbf{w}} = \big[1 - C(s)\big]\mathbf{w} + C(s)\big(\mathbf{w} - \hat{\mathbf{w}}\big). \tag{17}$$

Under Assumptions 1–2, the derivation in Appendix A gives

$$\|\tilde{\mathbf{w}}\| \le \begin{cases} \Delta_w, & 0 \le t < T_s, \\ L_w/\omega_c + (\zeta + \zeta_3)\,T_s, & t \ge T_s, \end{cases} \tag{18}$$

with $L_w := L_{w_t} + L_{w_\nu}\phi$ and $\zeta_3 := \Delta_w\,\omega_c$. Writing

$$\mu_w := L_w/\omega_c + (\zeta + \zeta_3)\,T_s \tag{19}$$

gives the ultimate bound $\|\tilde{\mathbf{w}}\| \le \mu_w$ after the transient $\Delta_w e^{-\omega_c t}$ decays. Since the DOB runs from the approach phase, this transient (time constant $1/\omega_c$) vanishes before the barriers bind, so μ_w acts as a pointwise bound in the safety-critical phase, used by the RaCBF margin and the robustness result.

C. DOB-MPC

The DOB-MPC is formulated to track the phase dependent reference while explicitly handling the actuator bounds and rate limits, and the safety constraints, with the disturbance estimate $\hat{d}$ embedded in the prediction model. The state and input vectors are $\mathbf{x} = [x, y, \psi, u, v, r]^\top$ and $\mathbf{u} = [n_S, n_T]^\top$. At each controller step t, the DOB-MPC solves the finite-horizon optimal control problem (OCP), formulated as:

$$\min_{\mathbf{u}(\cdot)} \sum_{k=0}^{N-1} \big(\|\mathbf{x}_{t+k} - \mathbf{r}_{t+k}\|_Q^2 + \|\mathbf{u}_{t+k}\|_R^2\big) + \|\mathbf{x}_{t+N} - \mathbf{r}_{t+N}\|_{Q_N}^2 \tag{20a}$$

$$\text{s.t.}\quad \mathbf{x}_t = \mathbf{x}(t), \tag{20b}$$

$$\mathbf{x}_{k+1} = f_d(\mathbf{x}_k, \mathbf{u}_k, \hat{d}_k), \tag{20c}$$

$$g(\mathbf{u}_k) \le 0, \tag{20d}$$

$$\Phi_j(\mathbf{x}_k, \mathbf{u}_k, \hat{d}_k) \ge 0, \quad j \in \mathcal{J}, \tag{20e}$$

where Q, R, and Q_N are positive-definite weighting matrices that penalize the state-tracking error, control effort, and terminal-tracking error, respectively, $\mathbf{r}_{t+k}$ is the phase-dependent reference, N is the prediction horizon, f_d is the discretized plant (1)–(2), (20d) imposes the actuator amplitude and rate limits as defined in (5) and (6), and Φ_j denotes the phase-specific RaCBF constraints derived in Section IV-D.

Let $\{\mathbf{u}_{t|t}, \ldots, \mathbf{u}_{t+N-1|t}\}$ denote the optimal input sequence solving (20) at time t. Only the first input $\mathbf{u}_{t|t}$ is applied to the plant, and the OCP is resolved at the next controller step with the updated state $\mathbf{x}(t)$ and disturbance estimate $\hat{d}_t$. The OCP is implemented using the sequential quadratic programming-real time iteration (SQP-RTI) algorithm in acados [33] with the HPIPM interior-point quadratic programming (QP) solver [34].

1) Robustness analysis: The OCP (20) couples a tracking objective (20a) with the safety constraints (20e), and we characterize the closed loop through two complementary properties of this single optimization rather than through one joint certificate. This subsection establishes a *robustness* property of the objective. The disturbance-aware tracking layer is closed-loop bounded, with a residual that scales with the DOB error bound μ_w. Section IV-D then establishes a *safety* property of the constraints. Any input satisfying the RaCBF condition renders the safe set forward invariant under the disturbance.

Each property holds under its own minimal assumptions. Their simultaneous satisfaction at every step, that is, recursive feasibility of the constrained nonlinear program, is in general an open problem for MPC with CBF constraints [30], [31], [35], [36] and is not claimed here but verified empirically over the Monte Carlo study of Section V.

We first establish the boundedness property for the tracking layer, that is, for (20) with the safety constraints (20e) inactive, so that only the input constraints (20d) remain. Let $V_t(\mathbf{x};\hat{d})$ denote the optimal value of (20) starting from $\mathbf{x}$ with the prediction model evaluated at the constant disturbance estimate $\hat{d}$, and let $c(\mathbf{x},\mathbf{r},\mathbf{u}) = \|\mathbf{x}-\mathbf{r}\|_Q^2 + \|\mathbf{u}\|_R^2$ be the stage cost in (20a). The argument follows the MPC framework of [37] extended to a time-varying disturbance estimate, and rests on two additional hypotheses beyond Assumptions 1–2.

Assumption 3 (Cost and value function bounds)**.** *There exist $\sigma \in \mathcal{K}_\infty$ and constants $0 < \underline{\lambda} \le \bar{\lambda}$ such that*

$$c(\mathbf{x},\mathbf{r},\mathbf{u}) \ge \underline{\lambda}\,\sigma(\mathbf{x}-\mathbf{r}), \quad V_t(\mathbf{x};\hat{d}) \le \bar{\lambda}\,\sigma(\mathbf{x}-\mathbf{r}),$$

for all $(\mathbf{x},\mathbf{r},\mathbf{u})$ in the feasible set of (20) *and all $\hat{d} \in \mathcal{D}$.*

For the stage cost (20a) with $Q \succ 0$, the positive-definite state penalty renders the cost detectable, so it lower-bounds a class-$\mathcal{K}_\infty$ measure σ of the tracking error, and the value-function upper bound with the same σ follows from asymptotic controllability of the plant on the compact feasible set [37, Sec. III]. Such cost and value-function bounds are standard in MPC stability analysis [6], [7], [38].

Assumption 4 (Lipschitz dynamics and value function)**.** *The plant f is locally Lipschitz in $\hat{d}$ with constant L_d, and the value function V_t is locally Lipschitz in its state and disturbance arguments with constants L_V and $L_{V,d}$, respectively.*

The plant is Lipschitz in $\hat{d}$ since $\hat{d}$ enters the kinetics (2) additively, and the value-function Lipschitz bounds follow from a maximum-theorem argument on the compact feasible set [39, Sec. C.3].

Theorem 1 (Closed-Loop Boundedness of the Tracking Layer)**.** *For the tracking layer* (20) *with the safety constraints* (20e) *inactive, under Assumptions 1–4 and N sufficiently large, with $C := L_V L_d \mu_w + L_{V,d}(2\mu_w + L_w T_c)$ and the contraction factor $\varepsilon \in (0,1)$ of Lemma 1,*

$$\sigma(\mathbf{x}_t - \mathbf{r}_t) \le \tfrac{\bar{\lambda}}{\underline{\lambda}}\,\varepsilon^{t-1}\,\sigma(\mathbf{x}_1 - \mathbf{r}_1) + \tfrac{C}{\underline{\lambda}(1-\varepsilon)}, \quad t \ge 1, \tag{21}$$

so $\mathbf{x}_t$ is ultimately bounded in the residual set

$$\mathcal{B} := \left\{\mathbf{x} \;:\; \sigma(\mathbf{x}-\mathbf{r}) \le \tfrac{C}{\underline{\lambda}(1-\varepsilon)}\right\}.$$

Here T_c denotes the controller sampling interval.

Proof. See Appendix B. ■

The right-hand side of (21) is the sum of a geometrically decaying transient and a constant residual, so the tracking error $\sigma(\mathbf{x}_t - \mathbf{r}_t)$ is driven into and stays within a neighborhood of size $C/[\underline{\lambda}(1-\varepsilon)]$ around the reference asymptotically. This residual shrinks linearly with μ_w and T_c, so a tighter DOB error bound and a faster controller cycle both contract the closed-loop bound.

D. Recovery RaCBF

Safe recovery imposes two geometric requirements: the USV must never collide with the mothership hull, and it must be guided into the cradle through the docking corridor. These are enforced by two phase-specific barrier functions, a side CBF and a funnel-shaped docking CBF, that enter the MPC as the safety constraints (20e) and are each robustified against the disturbance through the DOB estimate $\hat{\mathbf{d}}$ and its error bound μ_w. The following first reviews the nominal CBF condition, then derives the RaCBF form that each Φ_j takes, and finally instantiates it for the side and docking barriers.

1) Nominal CBF: Consider the nominal system $\dot{\mathbf{x}} = f(\mathbf{x},\mathbf{u})$ with safe set $\mathcal{C} = \{\mathbf{x} : B(\mathbf{x}) \ge 0\}$ defined as the zero-superlevel set of a continuously differentiable function B. Extending the control-affine formulation of [9] to nonlinear inputs, B is a *control barrier function* (CBF) if there exists $\alpha > 0$ such that for every $\mathbf{x} \in \partial\mathcal{C}$ some admissible $\mathbf{u}$ satisfies

$$\dot{B}(\mathbf{x},\mathbf{u}) + \alpha B(\mathbf{x}) = \tfrac{\partial B}{\partial \mathbf{x}}\, f(\mathbf{x},\mathbf{u}) + \alpha B(\mathbf{x}) \ge 0. \tag{22}$$

Any control law satisfying (22) renders $\mathcal{C}$ forward-invariant.

2) Robust Adaptive CBF: With environmental disturbances the true plant is $\dot{\mathbf{x}} = f(\mathbf{x},\mathbf{u}) + g_d(\mathbf{x})\mathbf{w}$, and the nominal CBF condition (22) must be strengthened. The time derivative of B along the disturbed system yields the disturbed CBF inequality

$$\tfrac{\partial B}{\partial \mathbf{x}}\, f(\mathbf{x},\mathbf{u}) + L_{g_d}B\,\mathbf{w} + \alpha B \ge 0. \tag{23}$$

where $L_{g_d}B := \frac{\partial B}{\partial \mathbf{x}}\, g_d(\mathbf{x})$ is the Lie derivative of B along g_d, through which the disturbance $\mathbf{w}$ enters.

Since $\mathbf{w}$ is unknown, (23) cannot be enforced directly. Decomposing $\mathbf{w} = \hat{d} + \tilde{\mathbf{w}}$ with $\|\tilde{\mathbf{w}}\| \le \mu_w$ from (18), the worst-case lower bound on $L_{g_d}B\,\mathbf{w}$ is

$$L_{g_d}B\,\mathbf{w} \ge L_{g_d}B\,\hat{d} - |L_{g_d}B|\,\mu_w. \tag{24}$$

Substituting into (23) yields the RaCBF constraint

$$\tfrac{\partial B}{\partial \mathbf{x}}\, f(\mathbf{x},\mathbf{u}) + \alpha B \;\ge\; -L_{g_d}B\,\hat{d} + |L_{g_d}B|\,\mu_w, \tag{25}$$

which depends only on the DOB estimate $\hat{d}$ from (12) and the error bound μ_w from (18). The constraint (25) is imposed at every prediction step inside the MPC as the safety constraint (20e).

3) Phase-specific barriers: The two safe sets, the half-space outside the mothership hull and the funnel-shaped corridor leading into the cradle, are encoded by the barrier functions shown in Fig. 2 (a).

All barrier functions are formulated in a cradle-fixed frame whose origin is attached to the cradle center. Because the mothership is far more massive than the USV, it advances at an approximately constant forward speed, and the cradle, mounted on the mothership, moves with it at the same speed, so the cradle-fixed frame translates at a nearly constant velocity and may be treated as inertial. For the same reason, the mothership's lateral and yaw responses to wave loading are slow and small, and the side cradle is additionally stabilized to suppress its residual lateral motion. Under these conditions the cradle-fixed frame is inertial and the cradle is stationary within it, so the barrier functions below are defined directly in this frame.

Side barrier : To prevent the USV from drifting into the mothership hull, define

$$h_{\text{side}}(\mathbf{x}) = y - y_{\text{MS}}, \tag{26}$$

where y is the USV lateral position and y_{MS} the mothership side boundary.

Funnel docking barrier : For a successful recovery, the USV's head point must cross the cradle threshold before its center of mass, so that once the head is inside the cradle the remainder of the hull follows along the corridor. The head point lies a distance l_h forward of the center of mass along the heading,

$$x_{hp} = x + l_h \cos\psi, \quad y_{hp} = y + l_h \sin\psi. \tag{27}$$

The head point (x_{hp}, y_{hp}) must therefore lie inside a funnel-shaped corridor that narrows to the cradle width. The terminal corridor width is set conservatively inside the physical cradle clearance, so that the enforced safe region retains a fixed lateral margin that absorbs the small residual motion of the actively stabilized cradle without relying on its exact magnitude. With head point coordinates relative to the cradle center (x_c, y_c),

$$x_r = x_{hp} - x_c, \quad y_r = y_{hp} - y_c, \tag{28}$$

the corridor boundary on each side is parameterized by

$$f_s(x_r) = a_s \tanh\big(b(x_0 - x_r)\big) + y_{0,s}, \qquad s \in \{\text{L}, \text{R}\}, \tag{29}$$

with shape parameters that may be chosen asymmetrically to fit the cradle and mothership geometry, and the left and right docking barriers are

$$h_{\text{dock,L}} = f_L(x_r) - y_r, \quad h_{\text{dock,R}} = f_R(x_r) + y_r. \tag{30}$$

The three barriers $h_{\text{side}}, h_{\text{dock,L}}, h_{\text{dock,R}}$ all have relative degree two with respect to both the input and the disturbance, since both enter the kinetics (2). The RaCBF tightening (25) requires the disturbance term to appear in $\dot{B}$, which is not the case for these barriers directly. We therefore adopt the HOCBF [10] construction: for each barrier h, define

$$\phi_1 = \dot{h} + \alpha_1 h, \quad \Phi := \dot{\phi}_1 + \alpha_2 \phi_1 \ \geq \ 0, \tag{31}$$

with $\alpha_1, \alpha_2 > 0$. The intermediate ϕ_1 has relative degree one with respect to both the input and the disturbance, both entering $\dot{\phi}_1$ through $\ddot{h}$, so applying (25) with $B = \phi_1$ yields the disturbance-robust constraints $\Phi_{\text{side}}, \Phi_{\text{dock,L}}, \Phi_{\text{dock,R}}$ imposed as the safety constraint (20e).

4) Robust forward invariance: The safety property announced in Section IV-C1 is a property of the RaCBF condition (25) itself. Each barrier h has relative degree two, so we form the HOCBF auxiliary $\phi_1 = \dot{h} + \alpha_1 h$ and start from a safe initial state with $h(\mathbf{x}_0) \geq 0$ and $\phi_1(\mathbf{x}_0) \geq 0$. Since the RaCBF constraint (25) with $B = \phi_1$ is tightened by the worst-case margin (24), enforcing it makes the nominal HOCBF inequality $\dot{\phi}_1 + \alpha_2 \phi_1 \geq 0$ hold along the true disturbed trajectory for every disturbance within $\|\mathbf{w} - \hat{d}\| \leq \mu_w$, which keeps ϕ_1 and in turn h nonnegative and thus renders the safe set $\mathcal{C} = \{\mathbf{x} : h(\mathbf{x}) \geq 0\}$ forward invariant. This is the relative-degree-two, additive-disturbance specialization of the HO-RaCBF forward-invariance result [11, Thm. 2] built on the HOCBF construction [10].

Physical Parameters		Hydrodynamic Parameters	
Parameter	**Value**	**Parameter**	**Value**
Length (L)	0.60 m	X_u	-0.026
Width (W)	0.25 m	$X_{u\|u\|}$	-0.006
Height (H)	0.20 m	Y_r	-0.046
m_{11}	1.98 kg	$Y_{v\|v\|}$	-0.957
m_{22}	1.10 kg	N_r	-0.033
m_{33}	0.062 kg m^2	N_v	-0.012
l_y	0.25 m	c	0.068
l_h	0.35 m		
α	5.24×10^{-3}		

TABLE I
PHYSICAL AND HYDRODYNAMIC PARAMETERS OF THE USV MODEL.

Remark 1. *The forward-invariance property above certifies the* condition*, not a particular controller. Any input that maintains* (25) *renders* $\mathcal{C}$ *invariant. What it does not establish is whether a given controller can maintain the condition at every step, that is, recursive feasibility of the RaCBF-constrained OCP* (20)*, which is generally not guaranteed for nonlinear MPC with CBF constraints [30], [31], [35], [36] and is not claimed here. Enforcing* (25) *across the prediction horizon rather than one step ahead aims to preserve feasibility by shaping the approach in advance, so that an admissible safe input more often remains available, as confirmed empirically by the Monte Carlo study of Section V.*

V. Numerical Simulation

The proposed framework is validated through two numerical simulation scenarios. The first scenario evaluates the trajectory tracking performance of the DOB-MPC in an approach and DP task without the RaCBF, isolating the disturbance-rejection capability of the DOB embedded in the MPC prediction model. In addition, the DOB estimation error is measured during the DP hold and compared against the analytical bound μ_w that the RaCBF uses in Scenario 2. The second scenario evaluates the overall recovery performance of the proposed framework on the complete two-phase recovery, including the tracking accuracy under disturbances and the safety contribution of the in-horizon RaCBF on top of the DOB-MPC. Both scenarios share the same USV model, controller settings, and disturbance generator. The following first introduces the simulation environment shared by both scenarios, then presents each scenario in turn.

The USV model used throughout this section is a 3-DOF maneuvering model identified from pool experiments in an indoor wave basin. The physical and hydrodynamic parameters of the identified model are summarized in Table I.

The controller is configured as follows. The DOB operates at 10 Hz ($T_s = 0.1$ s) using the piecewise-constant adaptation law (11) with adaptation gain $A_e = -I_3$ and low-pass cutoff frequency $\omega_c = 1.2$ rad/s. The MPC runs at 4 Hz ($T_c = 0.25$ s) with a prediction horizon of $N = 40$ steps ($T_f = 10$ s). The MPC cost weights in (20a) were empirically tuned to balance tracking accuracy and control effort, and are set to $Q = \text{diag}([10^3, 10^3, 10, 1, 1, 1])$, $Q_N = \text{diag}([10^4, 10^4, 10, 1, 1, 1])$, and $R = \text{diag}([1, 10])$ for the state, terminal, and input-rate penalties, respectively.

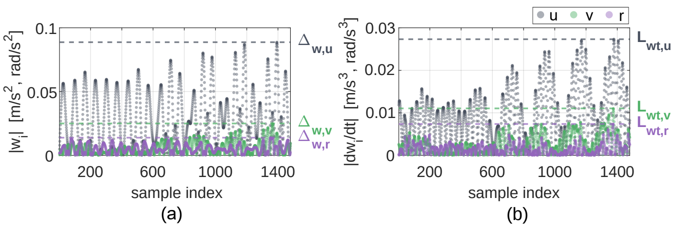

Fig. 4. Per-channel constants for the analytical bound, measured from the DP-hold run: (a) $|w_i|$ with the disturbance magnitude $\Delta_{w,i} = \max |w_i|$, and (b) $|\dot{w}_i|$ with the disturbance time rate $L_{w_t,i} = \max |\dot{w}_i|$.

Wind and wave disturbances are generated by the same model in both scenarios, following [32]. The wind load uses the relative-wind force model, and the wave load uses the first- and second-order wave load model combining a wave-frequency oscillatory component with a slowly-varying drift. The disturbance parameters correspond to Sea State 3 scaled to the $1/15$ wave-basin model, with wind speed, significant wave height, and peak wave period in $V_w \in [2.25,\, 3.10]$ m/s, $H_s \in [0.32,\, 0.58]$ m, and $T_p \in [4.0,\, 6.0]$ s, randomly sampled in both scenarios. Marine launch and recovery operations are typically performed with the mothership oriented into the prevailing wind and waves to minimize lateral disturbance during the maneuver. The wind and wave incidence angles are therefore sampled head-on within $\pm 60^\circ$ of the mothership heading.

The RaCBF margin is not hand-tuned but obtained per channel from the closed-form bound (19), so the tightening in (25) is applied channel-wise as $\sum_i |(L_{g_d}B)_i|\, \mu_{w,i}$. The disturbance is treated as exogenous in the state, in line with the framework convention of [23]. This is justified here because the dominant wave load is set by the external wave field, and because the recovery is executed at low speed, where the wind load varies with the vessel velocity only weakly through the relative-wind speed. The bound therefore depends only on the disturbance magnitude $\Delta_{w,i} := \max_t |w_i|$ and time rate $L_{w_t,i} := \max_t |\dot{w}_i|$ together with the DOB parameters ω_c, $|A_e|$, and T_s. These constants are obtained as the maxima over a preceding approach/DP run under head-on Sea-State-3 disturbance, shown in Fig. 4, yielding $\Delta_{w,u} = 0.089$, $\Delta_{w,v} = 0.025$, $\Delta_{w,r} = 0.014$ and $L_{w_t,u} = 0.027$, $L_{w_t,v} = 0.011$, $L_{w_t,r} = 0.007$. The resulting per-channel margins are $\mu_{w,u} = 0.049$ m/s^2, $\mu_{w,v} = 0.018$ m/s^2, and $\mu_{w,r} = 0.011$ rad/s^2, which the RaCBF (25) uses component-wise to tighten the barrier.

A. Scenario 1: Approach and Dynamic Positioning

The first scenario isolates the tracking layer of the proposed framework, quantifying how much the DOB contributes to disturbance rejection during the approach and DP hold. The RaCBF and the Phase-2 docking logic are therefore disabled, and the USV is commanded to acquire and hold a DP point located at the side of the mothership. The proposed DOB-MPC, a nominal MPC without disturbance compensation, and the adaptive line-of-sight guidance law (ALOS) of [40] are compared under identical, paired disturbance realizations.

Because the entire safety argument of Section IV-D rests on the disturbance estimate $\hat{\mathbf{d}}$ and its per-channel error bound $\mu_{w,i}$, we first verify that the DOB recovers the acting disturbance accurately during the approach and DP hold. Fig. 5 compares the true disturbance w_i with the DOB estimate $\hat{d}_i$ for each channel over a representative 40-s window of the settled regime. The DOB estimate tracks the environmental disturbance with a short lag across all three channels.

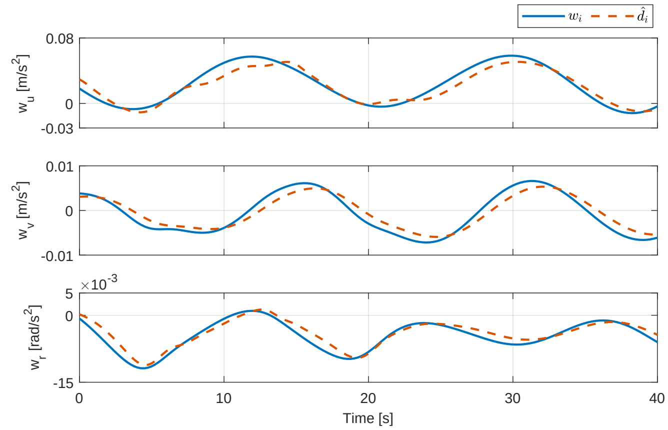

Fig. 5. Scenario 1 DOB tracking during the DP hold: true disturbance w_i (solid) versus the DOB estimate $\hat{d}_i$ (dashed) for the surge, sway, and yaw channels over a representative 40-s window.

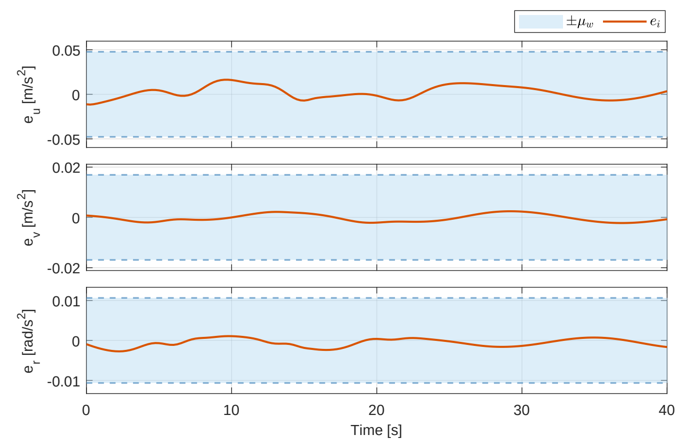

Fig. 6. Scenario 1 closed-loop estimation error $e_i = w_i - \hat{d}_i$ for the surge, sway, and yaw channels, shown with the analytical $\pm\mu_{w,i}$ band derived in (19). The error stays within the band throughout the window for every channel.

The corresponding closed-loop estimation error $e_i = w_i - \hat{d}_i$ is shown in Fig. 6 together with the per-channel $\pm\mu_{w,i}$ band derived in (19). The error stays well inside the band throughout the window. Over the full 400 s settled run the empirical worst-case error reaches 0.034 m/s^2, 0.007 m/s^2, and 0.006 rad/s^2 for the surge, sway, and yaw channels, all strictly below the corresponding analytical bounds with positive margin. This confirms that the analytical error bound is valid, so the DOB estimate can be trusted both for the disturbance-aware MPC prediction and for the RaCBF tightening used in Scenario 2.

Tracking accuracy is quantified by the root mean square error (RMSE) of the Euclidean distance error from the DP point, evaluated over a 50 s window after each controller first arrives within 0.5 m of the DP point, and averaged over 5 paired trials. Fig. 7 shows the distance error over a representative trial. The DOB-MPC drives the error to the smallest and tightest residual. The nominal MPC settles at a persistently

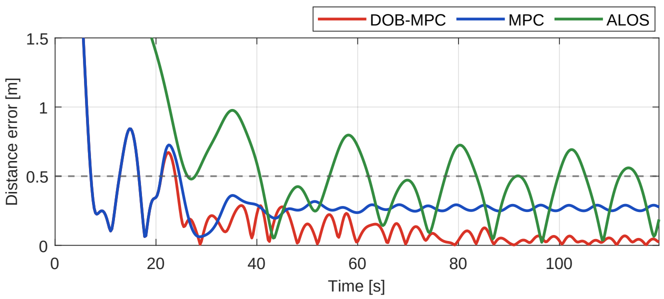

Fig. 7. Scenario 1 DP distance-tracking error over the evaluation window for a representative trial under wind and wave disturbance. The dashed line marks the 0.5 m arrival radius.

Controller	RMSE [m]
ALOS	0.667 ± 0.271
Nominal MPC	0.417 ± 0.085 ($37.5\,\%\,\blacktriangledown$)
DOB-MPC	$\mathbf{0.280 \pm 0.039}$ ($\mathbf{58.0\,\%\,\blacktriangledown}$)

TABLE II
SCENARIO 1 DP-TRACKING DISTANCE RMSE OVER A 50 S WINDOW AFTER ARRIVAL, AVERAGED OVER 5 PAIRED TRIALS.

larger offset, because without a disturbance estimate the MPC cannot cancel the sustained sway disturbance force and holds at a biased offset rather than converging to the DP point. The ALOS guidance exhibits the largest, persistently oscillatory error and the slowest convergence, because the ALOS structure does not model the vessel dynamics in which the disturbance acts, leaving it vulnerable to the time-varying wind and wave induced disturbance and unable to shape an effective approach and DP hold.

The aggregate tracking RMSE over the 5 trials is reported in Table II. The DOB-MPC attains the lowest RMSE of 0.280 m, a 32.9% reduction relative to the nominal MPC and a 58.0% reduction relative to the ALOS guidance, confirming that embedding the disturbance estimate in the prediction model is the dominant factor in DP tracking accuracy under environmental disturbance.

B. Scenario 2: Full Recovery

Scenario 2 evaluates the complete recovery of the USV into a cradle. To isolate the contribution of each of the three components of the proposed framework, the disturbance observer, the MPC tracking layer, and the RaCBF safety layer, the proposed DOB-MPC-CBF is compared against four baselines that ablate one component at a time. DOB-MPC removes the RaCBF constraints (20e) from the proposed framework, isolating the safety contribution of the CBF. MPC-CBF replaces the RaCBF with a nominal CBF and removes the DOB, isolating the contribution of the robust-adaptive robustification. ALOS-CBF replaces the MPC tracking layer with an ALOS augmented with the same DOB and the same RaCBF (25), but enforced one step ahead rather than across the MPC horizon. Because the throttle and steering enter the control forces nonlinearly and are coupled in (4), the one-step CBF condition is not affine in the input, so this safety filter is solved as a nonlinear program (NLP) rather than the quadratic program standard for control-affine systems, while still acting only one step ahead. ALOS is the bare ALOS guidance law without DOB or CBF, serving as the lower baseline. The two in-horizon CBF controllers, DOB-MPC-CBF and MPC-CBF, share the identical MPC and the same HOCBF gains $\alpha_{\mathrm{side},1} = \alpha_{\mathrm{side},2} = 0.7$, $\alpha_{\mathrm{dock},1} = 1.5$, $\alpha_{\mathrm{dock},2} = 1.3$. The funnel corridor (29) uses the shared parameters $b = 1.5$, $x_0 = -1.4$ m, with the mothership-side boundary $a_L = 0.20$, $y_{0,L} = 0.40$ m and the open-side boundary $a_R = 0.35$, $y_{0,R} = 0.55$ m, and the side boundary is placed at $y_{\mathrm{MS}} = -0.6$ m relative to the cradle center. DOB-MPC-CBF and ALOS-CBF share the same per-channel margins μ_w derived in the setup above, applied component-wise to tighten the barrier.

A Monte Carlo study of 100 trials is conducted to assess recovery capability under randomized conditions. Each trial randomizes the initial USV position within a radius of up to 3.5 m of a nominal start point with an initial heading offset of $-20°$, and applies the same disturbance realization to all five controllers within a trial so that the comparison is paired. A trial is counted as a success if the USV head point enters the cradle while the side barrier h_{side} and the funnel docking barriers $h_{\mathrm{dock,L}}$, $h_{\mathrm{dock,R}}$ all stay non-negative throughout the maneuver. A trial in which any barrier is breached is counted as a safety violation, separated into a side violation or a docking violation according to which barrier is crossed first.

Before turning to the aggregate statistics, Fig. 8 illustrates a representative successful recovery of the proposed DOB-MPC-CBF. The left panel shows the cradle-fixed trajectory. The USV is driven through the approach and DP phase to the DP point, and on the dwell-time transition enters the cradle along the funnel corridor, with the hull pose drawn every 3 s. The right panels show the corresponding side barrier h_{side} and the funnel docking barriers $h_{\mathrm{dock,L}}, h_{\mathrm{dock,R}}$ activated upon Phase 2 entry. The trajectory threads the funnel cleanly, and both barriers remain strictly positive throughout the maneuver, with the in-horizon RaCBF holding a comfortable margin from zero as the USV ingresses.

The aggregate results over the 100 trials are summarized in Fig. 9, and they separate the contribution of each of the three components. The barrier is the decisive factor for safe recovery. Removing it from the MPC track (DOB-MPC) collapses the success rate and produces frequent side and docking violations because the controller never reasons about the safe set, and the same effect appears on the ALOS track, where adding the barrier (ALOS-CBF) substantially lifts the success rate over the bare ALOS. Given the barrier, the disturbance compensation is what makes it reliable. MPC-CBF uses the same MPC and barrier as the proposed controller, but with the nominal CBF in place of the DOB-supplied margin, so the comparison isolates the robust-adaptive tightening alone. Adding that tightening, that is, turning the nominal barrier into a RaCBF, raises the success rate, eliminates the side-boundary violations. Because the DOB-supplied margin correctly informs the prediction model and tightens the barrier against the residual disturbance, the USV holds the corridor and a safe ingress remains feasible, whereas the nominal CBF must contend with the uncompensated disturbance and breaches or stalls more often. Finally, the way the barrier is enforced matters. Among the RaCBF controllers the DOB-MPC-CBF embeds the barrier across the MPC prediction horizon and markedly outperforms the one-step ALOS-CBF,

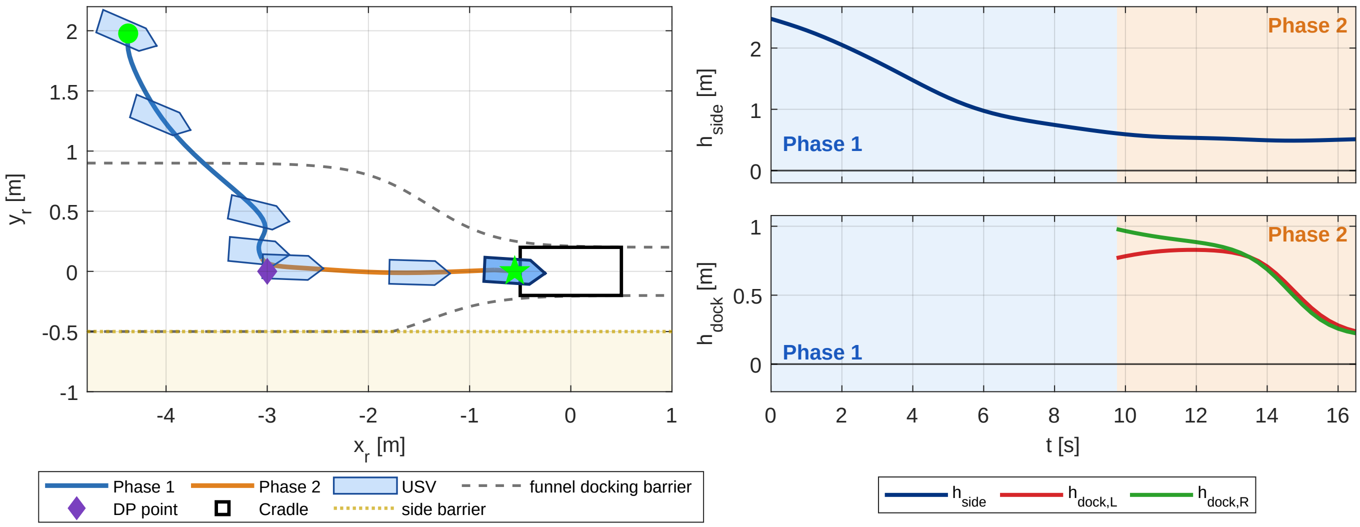

Fig. 8. Representative Scenario 2 recovery of the proposed DOB-MPC-CBF (cradle-fixed frame). Left: trajectory threading the funnel corridor (dashed) past the mothership side boundary y_{MS} (dotted) into the cradle (rectangle), with Phase 1/Phase 2 distinguished by color, the DP point marked, and the USV hull drawn every 3 s. Right: side barrier h_{side} (top) and funnel docking barriers $h_{\mathrm{dock,L}}, h_{\mathrm{dock,R}}$ (bottom, active in Phase 2), with $h_{\mathrm{dock,L}}$ shown in red and $h_{\mathrm{dock,R}}$ in green.

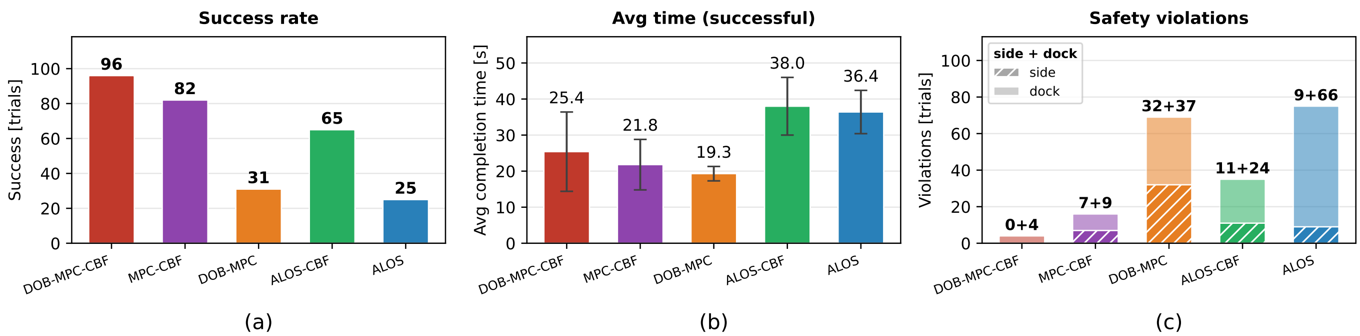

Fig. 9. Monte Carlo summary over 100 trials: (a) number of successful recoveries, (b) mean completion time over the successful trials, and (c) safety violations split into side-boundary (hatched) and docking-corridor (solid) breaches.

a gap whose mechanism is illustrated by the results. The MPC-based controllers also complete successful recoveries faster than the ALOS-based ones, as shown in Fig. 9 (b), because the MPC computes an optimal trajectory toward the cradle that accounts for the vessel dynamics and the estimated disturbance, achieving a more effective ingress than the ALOS guidance.

The per-step solver time, summarized in Table III, confirms that the proposed controller is real-time feasible, where all simulations were executed on a laptop with an Intel Core i7-1260P processor (2.1 GHz) and 32 GB of RAM. The MPC-based controllers run comfortably within the control period of the loop, with a worst case that never approaches the real-time limit over the entire study. Embedding the in-horizon RaCBF on top of DOB-MPC adds only a marginal increment to the tracking solve, so propagating safety across the prediction horizon is nearly free relative to the optimization itself. The one-step ALOS-CBF filter is cheaper than the MPC solves but, being an NLP in the nonlinear input, remains far more costly than the closed-form bare ALOS. The key observation is that the large safety improvement of the in-horizon RaCBF over the one-step filter is obtained at a comparable, real-time-feasible computational cost.

Although DOB-MPC-CBF and ALOS-CBF enforce the

Controller	Mean $\bar{t}_c$ [ms]	Max $t_c^{\max}$ [ms]
DOB-MPC-CBF	**20.8**	**33.4**
MPC-CBF	20.4	39.3
DOB-MPC	19.2	50.1
ALOS-CBF	9.5	43.4
ALOS	0.1	0.5

TABLE III
PER-STEP SOLVER TIME OVER THE 100 MONTE CARLO TRIALS, TO BE COMPARED AGAINST THE 250 MS CONTROL PERIOD OF THE 4 HZ LOOP.

same RaCBF with the same DOB and the same per-channel $\mu_{w,i}$ margins, the proposed in-horizon controller still attains a markedly higher success rate than the one-step ALOS-CBF, and this remaining gap stems from how each enforces forward invariance. A one-step RaCBF certifies safety by requiring a safe control input to exist at the current instant, which presumes sufficient control authority to keep the state inside the safe set. Under the relatively large disturbance and the underactuated dynamics of the USV this assumption can fail, so that the set of safe control inputs may become empty as the maneuver progresses, a limitation studied for input-constrained CBFs in [28], [29]. Embedding the RaCBF across the prediction horizon relaxes this requirement, because the MPC reasons about safety over the entire horizon rather than one step ahead and can shape the approach in advance so that a safe input remains available.

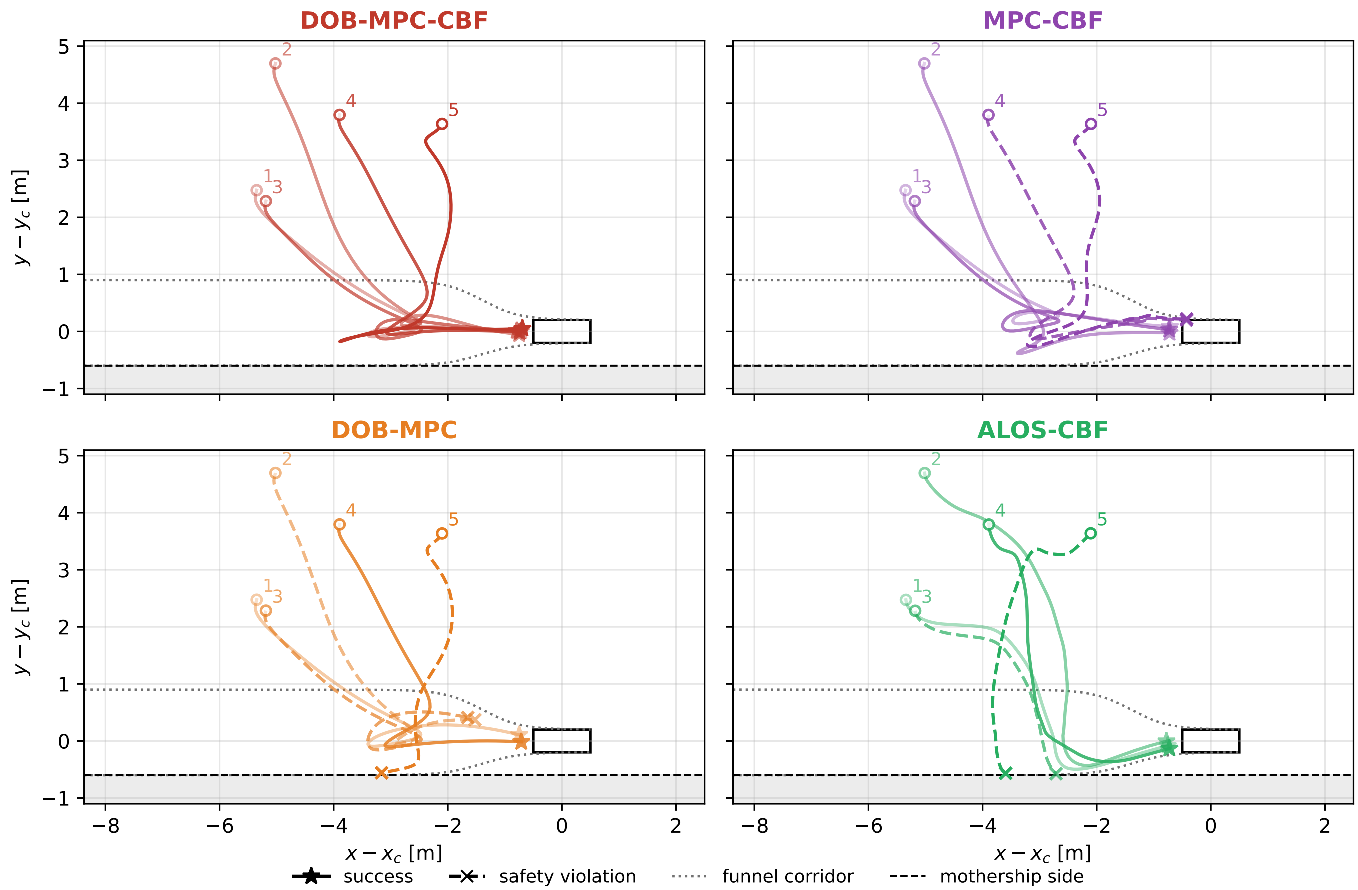

Fig. 10. Scenario 2 qualitative comparison in the cradle-fixed frame for four controllers (the bare ALOS omitted), drawn from the 100-trial study. The same five trials are shown in every panel, shaded light-to-dark by run so that a given run can be matched across controllers. Solid line with $\star$: success. Dashed line with $\times$: safety violation.

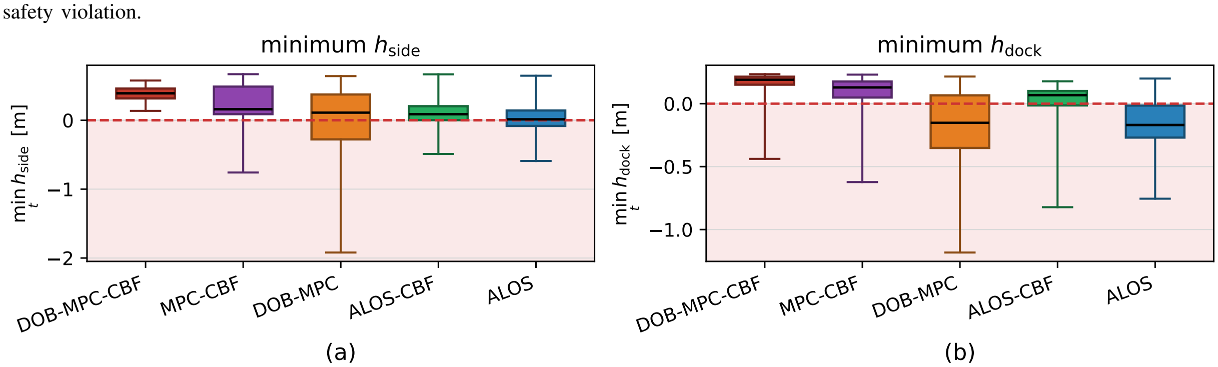

Fig. 11. Distribution of the minimum barrier values over the 100 Monte Carlo trials: (a) side barrier h_{side} and (b) docking barrier $h_{\text{dock}} = \min(h_{\text{dock,L}}, h_{\text{dock,R}})$. The box spans the interquartile range and the whiskers the full range across trials. Values below the red line lie in the unsafe region. The proposed DOB-MPC-CBF shows the tightest distribution and the highest barrier margin among all controllers.

To examine the behavior behind these aggregate numbers, Fig. 10 shows the four controllers on the same five trials, with the bare ALOS omitted and each run shaded light-to-dark so that a given run can be traced across panels. Run 1 is an easy case recovered by all four controllers. Run 2 isolates the CBF: the DOB-MPC without a barrier breaches the docking corridor while the three CBF-equipped controllers thread it. Run 3 isolates the tracking layer: the one-step ALOS-CBF cuts across the side boundary while the in-horizon MPC-based controllers hold the corridor. Run 4 isolates the robust-adaptive robustification: the nominal MPC-CBF breaches the docking corridor, whereas the proposed DOB-MPC-CBF, whose barrier is tightened by the DOB-supplied margin, threads it. Run 5 is the hardest case, in which every baseline is forced across a boundary and only the proposed DOB-MPC-CBF threads the cradle. The proposed controller recovers successfully in every run shown, while across the panels the robust tracking layer governs how smoothly the USV follows the funnel and the in-horizon RaCBF governs how tightly the trajectories stay inside the safe corridor near the boundary.

The safety margin behind these success rates is quantified in Fig. 11, which reports the distribution of the minimum side and docking barrier values over all 100 trials. The proposed DOB-MPC-CBF holds both barriers within the tightest, highest band

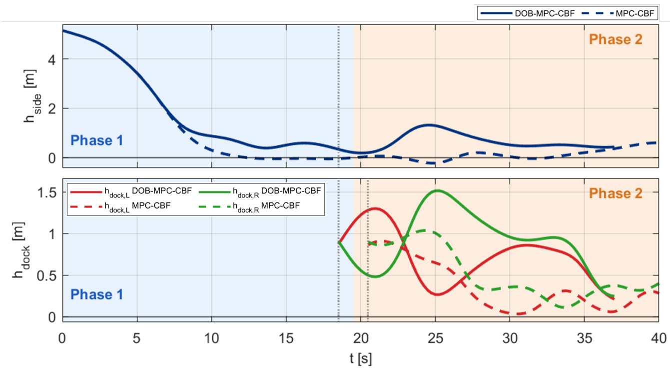

Fig. 12. CBF comparison ($V_w = 4.76$ m/s, $H_s = 0.40$ m). Top: side barrier h_{side}. Bottom: docking barrier $h_{\mathrm{dock,L}}, h_{\mathrm{dock,R}}$, Phase 2 only. The proposed DOB-MPC-CBF (solid blue) holds both barriers strictly positive and completes the recovery, while the nominal MPC-CBF (dashed red) breaches the side barrier under the same disturbance realization.

of all controllers, rarely crossing either boundary, whereas the no-CBF controllers and the one-step ALOS-CBF filter let the barriers spread well into the unsafe region. This distribution reflects what each controller is missing relative to the proposed one. DOB-MPC and the bare ALOS carry no barrier and cross the boundary outright. MPC-CBF enforces a barrier but, without the DOB-supplied margin, cannot keep it robust to the disturbance, and the one-step ALOS-CBF cannot plan the constrained maneuver across the horizon, so both let the barrier cross zero. Only the proposed controller combines the barrier, the disturbance compensation that robustifies it, and the horizon planning that keeps it feasible.

The mechanism by which the DOB-supplied tightening converts the nominal CBF into a robust one is illustrated in Fig. 12, which compares the barrier evolution of the proposed DOB-MPC-CBF and the nominal MPC-CBF on a harsh trial with $V_w = 4.76$ m/s and $H_s = 0.40$ m. Both controllers face the same disturbance realization with the same MPC and barrier formulation, the only difference being the DOB-supplied tightening in DOB-MPC-CBF. The side barrier of MPC-CBF descends below zero shortly after the funnel approach, reaching a minimum of -0.23 m that corresponds to a hull-contact breach, whereas the DOB-MPC-CBF side barrier holds at a comfortable positive margin with a minimum of $+0.18$ m. Once Phase 2 is reached, the docking barrier of MPC-CBF oscillates at the safety threshold with a minimum of $+0.04$ m while DOB-MPC-CBF holds it at $+0.21$ m. This single-trial view illustrates the mechanism behind the aggregate distribution. Without the DOB-supplied margin, the nominal CBF cannot robustify the barrier against the residual disturbance, so the boundary is breached under the same condition where the proposed framework completes the recovery safely.

Together, the aggregate ablation, the barrier-margin distribution, and the trajectory-level mechanism show that the DOB, the in-horizon RaCBF, and the robust MPC tracking layer are jointly necessary, and that their combination converts robust tracking into a reliable and safe recovery.

VI. Pool Experiment

The proposed framework was further validated through a pool experiment in an indoor wave basin, complementing the numerical simulation. The experimental setup is shown in Fig. 13. A wave generator installed at one end of the basin produced random irregular waves following the JONSWAP spectrum scaled to Sea State 3 on the $1/15$ basin scale, with the same significant wave height range $H_s \in [0.32,\, 0.58]$ m used in Section V. Because the mothership-cradle structure is moored at the center of the basin by spring cables and cannot move forward, an array of current generators produced a steady 0.3 to 0.5 m/s inflow along the mothership axis, so that the USV experiences the same relative water flow it would encounter if the mothership were advancing forward during a real recovery. The cradle was actively stabilized to suppress its lateral motion under wave loading, providing a laterally steady recovery target throughout the run. The USV is a $1/15$ scaled monohull vessel actuated by a single stern outboard engine, and its pose was measured in real time by an OptiTrack motion capture system through IR LED markers fitted on its deck.

The DOB, MPC, and RaCBF weights and parameters were re-tuned for the pool experiment to reflect the actuator response, sensor noise, and unmodeled hydrodynamic effects of the physical USV. The DOB operates at 10 Hz ($T_s = 0.1$ s) with adaptation gain $A_e = -I_3$ and low-pass cutoff frequency $\omega_c = 0.5$ rad/s, and the MPC runs at 4 Hz ($T_c = 0.25$ s) with a prediction horizon of $N = 40$ steps ($T_f = 10$ s). The MPC cost weights are set to $Q = \mathrm{diag}([10^3,\, 10^3,\, 10^2,\, 10,\, 10^{-1},\, 1])$, $Q_N = \mathrm{diag}([10^4,\, 10^4,\, 10,\, 1,\, 1,\, 1])$, and $R = \mathrm{diag}([10^{-1},\, 10^2])$. Each barrier is enforced as a relative-degree-two HOCBF with a pair of gains (α_1, α_2) as in (31), where the side barrier uses $\alpha_{\mathrm{side},1} = \alpha_{\mathrm{side},2} = 0.75$ and the docking barriers use $\alpha_{\mathrm{dock},1} = 1.0$ and $\alpha_{\mathrm{dock},2} = 1.2$. The disturbance margin $\mu_{w,u} = 0.075$ m/s^2, $\mu_{w,v} = 0.031$ m/s^2, and $\mu_{w,r} = 0.021$ rad/s^2 were computed from the closed-form estimation-error bound (19), evaluated with the disturbance regularity constants derived from the model identified during the dynamic-positioning experiment, so that the margin reflects the disturbance characteristics actually observed in the basin. The funnel docking corridor (29) uses the same shape parameters as in the simulation.

Two scenarios were conducted, mirroring the two simulation scenarios.

A. Scenario 1: Approach and Dynamic Positioning

The first scenario evaluates the tracking layer by commanding the USV to hold the DP standoff point off the cradle under the basin disturbance, with no safety constraint active. Three tracking controllers were compared on hardware, namely a baseline ALOS guidance law, a nominal MPC without disturbance compensation, and the proposed DOB-MPC. Each controller was run for 50 s over two paired trials. Fig. 14 shows the time history of the Euclidean position error from the DP target. All three controllers exhibit oscillatory error driven by the wave encounter, but the DOB-MPC maintains a visibly lower error envelope throughout the run, whereas the

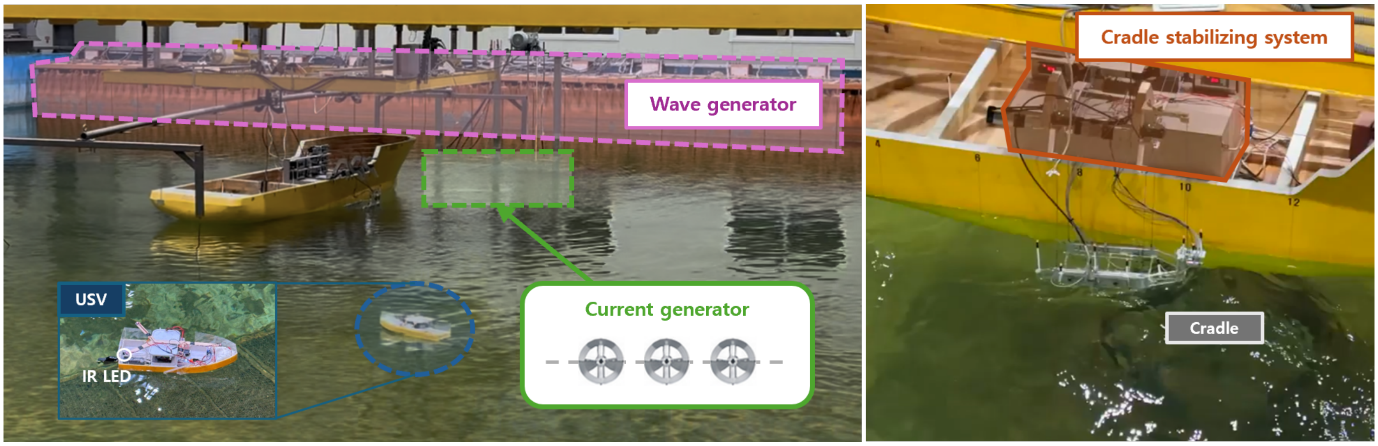

Fig. 13. Pool experimental setup. The indoor wave basin is equipped with a wave generator and submerged current generators (left), and the side-mounted cradle is attached to the basin sidewall through a stabilizing mechanism (right). The USV is tracked by an overhead optical system through IR LED markers.

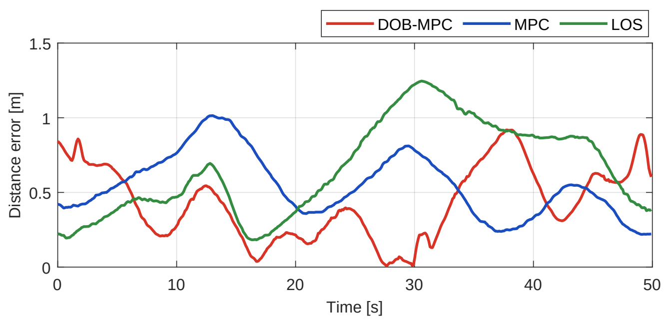

Fig. 14. Time history of the Euclidean position error from the DP standoff point under the basin disturbance in pool DP scenario.

Controller	Trial 1 [m]	Trial 2 [m]	Average [m]
ALOS	0.5964	0.6394	0.6179
Nominal MPC	0.6332	0.5269	0.5801 (6.1% ▼)
DOB-MPC	**0.4442**	**0.4459**	**0.4451 (28.0% ▼)**

TABLE IV
POOL DP SCENARIO TRACKING RMSE OVER A 50 S WINDOW FOR TWO PAIRED TRIALS.

ALOS and the nominal MPC drift further from the standoff point during disturbance peaks.

The quantitative summary in Table IV confirms this trend, with the DOB-MPC achieving the lowest tracking RMSE among the three controllers and improving substantially on both the nominal MPC and the ALOS baseline. As in the simulation of Section V, the gap between the DOB-MPC and the nominal MPC isolates the contribution of the DOB, so the pool result confirms that the disturbance-rejection benefit of embedding the estimate in the prediction model transfers from simulation to physical hardware.

B. Scenario 2: Full Recovery

The second scenario deploys the complete two-phase recovery framework, in which the USV starts from a position 7 m off the starboard quarter of the mothership and executes the full sequence of approach, DP, transition to the docking phase, and ingress into the cradle through the funnel corridor, under the same basin disturbance as the first scenario. Only the proposed DOB-MPC-CBF is deployed in the pool, since the three baselines all lack in-horizon RaCBF enforcement and would risk physical collision with the mothership and cradle hardware, the failure mode that the RaCBF is designed to prevent. The pool recovery scenario therefore demonstrates the physical feasibility of the proposed framework on hardware, completing a full cradle recovery without collision. Two representative trials drawn from the pool campaign are presented to illustrate this behavior.

Fig. 15. Snapshots of the USV executing the proposed DOB-MPC-CBF recovery into the side-mounted cradle.

Snapshots of the full recovery process in Fig. 15 show the USV over the entire sequence, from the approach and DP hold through the funnel ingress into the cradle. The realized trajectories of two independent recovery trials are overlaid in Fig. 16 on the mothership side boundary, the funnel docking corridor, and the cradle, with the corresponding barrier time histories in Fig. 17. Across both trials, the lateral position remains clear of the mothership side boundary so that h_{side} stays strictly positive throughout the run, and once the docking phase is activated, the funnel docking barrier contracts toward zero from above without crossing it. The consistency of this safety-critical behavior across the two independent trials indicates that the disturbance-robust tracking of the DOB-MPC and the forward-invariance property enforced by the in-horizon RaCBF constraints are both preserved on physical hardware under the basin disturbance.

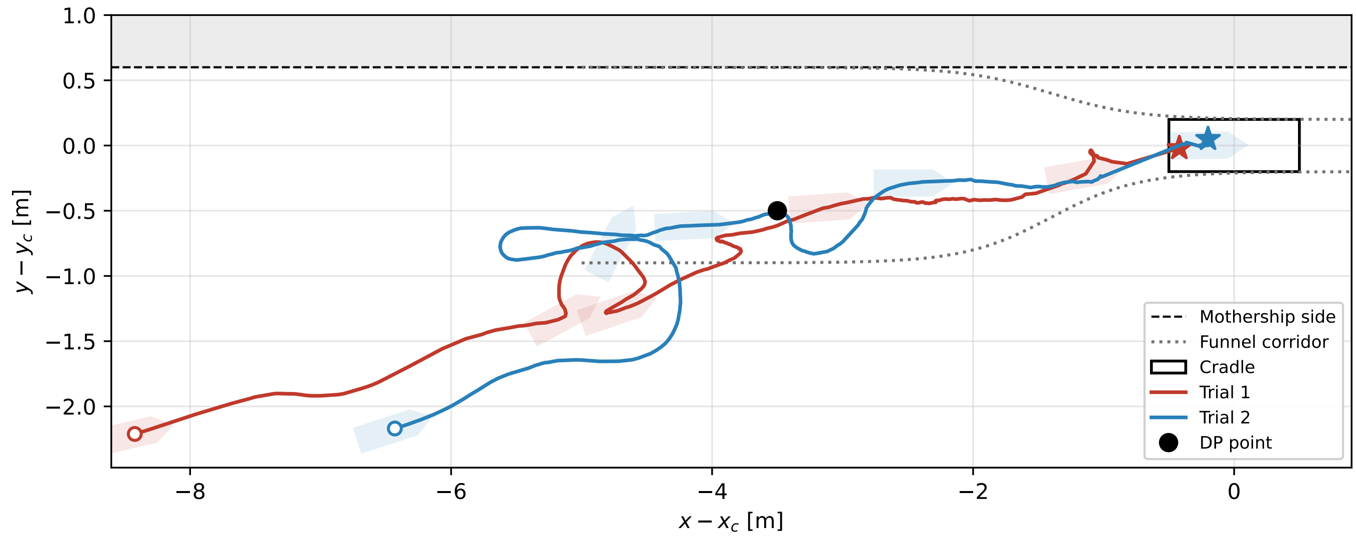

Fig. 16. Pool full recovery trajectories of the proposed DOB-MPC-CBF for two independent trials in the cradle-fixed frame, overlaid on the mothership side boundary (dashed, with the unsafe region shaded), the funnel docking corridor (dotted), and the cradle (rectangle). The marker ○ denotes the start of each trial and ⋆ the cradle entry.

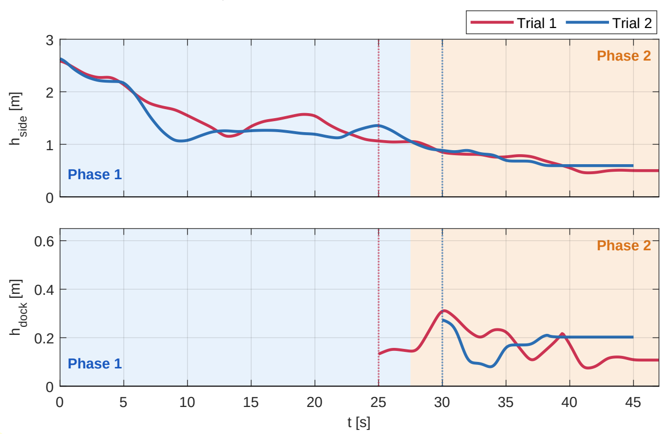

Fig. 17. Pool full recovery time histories of the side CBF h_{side} (top) and the funnel docking CBF $h_{\mathrm{dock}} = \min(h_{\mathrm{dock,L}}, h_{\mathrm{dock,R}})$ (bottom) for the two trials of Fig. 16. The dotted vertical line marks the transition to the docking phase for each trial.

VII. Conclusion

This paper presented a safe autonomous recovery framework that combines DOB-MPC and phase-specific RaCBFs for docking an underactuated USV into a side-mounted cradle under environmental disturbances. The DOB-MPC embeds a disturbance estimate into the prediction model for robust tracking, supported by a closed-form estimation-error bound and a closed-loop boundedness guarantee. A side-boundary barrier and a funnel-shaped docking barrier, tightened online by this bound and enforced across the MPC horizon, secure the two-phase recovery. Monte Carlo simulations and pool experiments under scaled environmental disturbances confirmed reliable, collision-free recovery with the highest success rate among the compared controllers, showing that the robust tracking layer and the in-horizon safety layer are jointly necessary.

Appendix A
Derivation of the Filtered Estimation-Error Bound

We derive (18) from (17) by bounding the two terms separately.

Filtering bias $\mathbf{e}_b := [1 - C(s)]\mathbf{w}$.*:* Integration by parts on $C(s)\mathbf{w}(t) = \int_0^t \omega_c e^{-\omega_c(t-\tau)}\mathbf{w}(\tau)d\tau$ gives

$$\mathbf{e}_b(t) = \mathbf{w}(0)e^{-\omega_c t} + \int_0^t e^{-\omega_c(t-\tau)}\dot{\mathbf{w}}(\tau)\,d\tau. \tag{32}$$

Since $\|\dot{\mathbf{w}}\| \le L_{w_t} + L_{w_\nu}\|\dot{}$ $\nu\| \le L_{w_t} + L_{w_\nu}\phi =: L_w$ by Assumptions 1–2,

$$\|\mathbf{e}_b(t)\| \le \Delta_w\, e^{-\omega_c t} + \tfrac{L_w}{\omega_c}\big(1 - e^{-\omega_c t}\big). \tag{33}$$

Filtered raw error $\mathbf{e}_r := C(s)(\mathbf{w} - \hat{\mathbf{w}})$.*:* The Laplace inverse of $\mathbf{e}_r(s)$ obeys $\dot{\mathbf{e}}_r = -\omega_c\mathbf{e}_r + \omega_c(\mathbf{w} - \hat{\mathbf{w}})$ with $\mathbf{e}_r(0) = 0$. For $t \in [0, T_s)$, $\hat{\mathbf{w}} = 0$ and $\|\mathbf{w}\| \le \Delta_w$, so

$$\|\mathbf{e}_r(t)\| \le \Delta_w\big(1 - e^{-\omega_c t}\big) \le \zeta_3 T_s, \tag{34}$$

with $\zeta_3 := \Delta_w\omega_c$. For $t \ge T_s$, splitting the solution at T_s and applying (13),

$$\|\mathbf{e}_r(t)\| \le e^{-\omega_c(t-T_s)}\zeta_3 T_s + \big(1 - e^{-\omega_c(t-T_s)}\big)\zeta T_s \le (\zeta + \zeta_3)T_s. \tag{35}$$

Combined bound.: For $0 \le t < T_s$, $\hat{d} = 0$ gives the direct bound $\|\tilde{\mathbf{w}}\| = \|\mathbf{w}\| \le \Delta_w$. For $t \ge T_s$, the triangle inequality with (33) and (35) yields

$$\|\tilde{\mathbf{w}}(t)\| \le L_w/\omega_c + (\zeta + \zeta_3)T_s + \Delta_w e^{-\omega_c t}. \tag{36}$$

The transient $\Delta_w e^{-\omega_c t}$ decays exponentially, so $\|\tilde{\mathbf{w}}\| \le \mu_w$ holds as the ultimate bound, establishing (18).

Appendix B
Proof of the Closed-Loop Boundedness

This appendix establishes Theorem 1 by first proving Lemma 1 and Lemma 2, and then combining them.

Lemma 1. *Under Assumption 3, for N sufficiently large there exists $\varepsilon \in (0,1)$ such that with the disturbance estimate held constant at $\hat{d}_t$ and $\hat{\mathbf{x}}_{t+1} := f_d(\mathbf{x}_t, \mathbf{u}_{t|t}, \hat{d}_t)$,*

$$V_{t+1}(\hat{\mathbf{x}}_{t+1}; \hat{d}_t) \le \varepsilon\, V_t(\mathbf{x}_t; \hat{d}_t). \tag{37}$$

Proof of Lemma 1. Let $\{\hat{\mathbf{x}}_{t|t}, \hat{\mathbf{x}}_{t+1|t}, \dots, \hat{\mathbf{x}}_{t+N|t}\}$, with $\hat{\mathbf{x}}_{t|t} = \mathbf{x}_t$, be the optimal predicted trajectory at t under the constant estimate $\hat{d}_t$, and abbreviate $\sigma_k := \sigma(\hat{\mathbf{x}}_{t+k|t} - \mathbf{r}_{t+k})$ with $\sigma_0 = \sigma(\mathbf{x}_t - \mathbf{r}_t)$, $c_k := c(\hat{\mathbf{x}}_{t+k|t}, \mathbf{r}_{t+k}, \mathbf{u}_{t+k|t})$, and $c_0 := c(\mathbf{x}_t, \mathbf{r}_t, \mathbf{u}_{t|t})$. The argument specializes the horizon-length contraction of [37] to the quadratic cost (20a), for which $\sigma(\mathbf{x} - \mathbf{r})$ is detectable from c trivially through the lower bound of Assumption 3.

The per-stage lower bound $c_k \ge \underline{\lambda}\,\sigma_k$ of Assumption 3 and the nonnegativity of the terminal cost give $V_t(\mathbf{x}_t; \hat{d}_t) = \sum_{k=0}^{N-1} c_k + \|\hat{\mathbf{x}}_{t+N|t} - \mathbf{r}_{t+N}\|^2_{Q_N} \ge \underline{\lambda} \sum_{k=0}^{N-1} \sigma_k$, which with the upper bound $V_t(\mathbf{x}_t; \hat{d}_t) \le \bar{\lambda}\,\sigma_0$ of Assumption 3 and the removal of the nonnegative $k = 0$ term yields $\sum_{k=1}^{N-1} \sigma_k \le (\bar{\lambda}/\underline{\lambda})\,\sigma_0$. As the $N-1$ terms are nonnegative, at least one interior index $k^\star \in \{1, \dots, N-1\}$ satisfies

$$\sigma_{k^\star} \le \frac{1}{N-1} \frac{\bar{\lambda}}{\underline{\lambda}}\,\sigma_0 = \frac{\kappa_0}{N-1}\,\sigma_0, \qquad \kappa_0 := \frac{\bar{\lambda}}{\underline{\lambda}}. \tag{38}$$

Since $V_{t+1}(\hat{\mathbf{x}}_{t+1}; \hat{d}_t)$ is the optimal value, it is bounded above by the cost of any admissible candidate at $t+1$. Reuse the first $k^\star - 1$ optimal inputs $\{\mathbf{u}_{t+1|t}, \dots, \mathbf{u}_{t+(k^\star-1)|t}\}$, which carry the nominal successor $\hat{\mathbf{x}}_{t+1} = \hat{\mathbf{x}}_{t+1|t}$ to the cheap node $\hat{\mathbf{x}}_{t+k^\star|t}$, and append an admissible block of length $j := N - k^\star + 1$ from it, for a total of N inputs. As the tracking layer carries only the input constraints (20d) this candidate is admissible, and since the constant-estimate model and the shared start $\hat{\mathbf{x}}_{t+1|t}$ make the reused stages identical to those of the t-solution, suboptimality gives

$$V_{t+1}(\hat{\mathbf{x}}_{t+1}; \hat{d}_t) \le \sum_{k=1}^{k^\star - 1} c_k + V^{(j)}(\hat{\mathbf{x}}_{t+k^\star|t}; \hat{d}_t), \tag{39}$$

where $V^{(j)}$ is the optimal value of the j-step OCP. The reused stages are part of the t-cost, so $\sum_{k=1}^{k^\star-1} c_k \le V_t(\mathbf{x}_t; \hat{d}_t) - c_0$, while the appended cost-to-go is bounded through the value-function upper bound of Assumption 3 and (38) by

$$V^{(j)}(\hat{\mathbf{x}}_{t+k^\star|t}; \hat{d}_t) \le \bar{\lambda}\,\sigma_{k^\star} \le \delta_N\,\sigma_0, \qquad \delta_N := \frac{\bar{\lambda}\,\kappa_0}{N-1}. \tag{40}$$

Substituting both into (39) and applying $c_0 \ge \underline{\lambda}\,\sigma_0$,

$$V_{t+1}(\hat{\mathbf{x}}_{t+1}; \hat{d}_t) \le V_t(\mathbf{x}_t; \hat{d}_t) - (\underline{\lambda} - \delta_N)\,\sigma_0. \tag{41}$$

The upper bound of Assumption 3 gives $\sigma_0 \ge V_t(\mathbf{x}_t; \hat{d}_t)/\bar{\lambda}$, so for N large enough that $\delta_N < \underline{\lambda}$ the positive coefficient $\underline{\lambda} - \delta_N$ turns (41) into

$$V_{t+1}(\hat{\mathbf{x}}_{t+1}; \hat{d}_t) \le \left[1 - \tfrac{\underline{\lambda} - \delta_N}{\bar{\lambda}}\right] V_t(\mathbf{x}_t; \hat{d}_t), \tag{42}$$

which is (37) with $\varepsilon = 1 - (\underline{\lambda} - \delta_N)/\bar{\lambda} \in (0,1)$. The condition $\delta_N < \underline{\lambda}$ holds whenever $N > 1 + (\bar{\lambda}/\underline{\lambda})^2 = 1 + \kappa_0^2$. ■

Lemma 2. *Under Assumption 4, let the actual and predicted states at $t+1$ be $\mathbf{x}_{t+1} = f_d(\mathbf{x}_t, \mathbf{u}_{t|t}, \mathbf{w}_t)$ and $\hat{\mathbf{x}}_{t+1} = f_d(\mathbf{x}_t, \mathbf{u}_{t|t}, \hat{d}_t)$, respectively. Then the following inequality holds*

$$\begin{aligned} V_{t+1}(\mathbf{x}_{t+1}; \hat{d}_{t+1}) - V_{t+1}(\hat{\mathbf{x}}_{t+1}; \hat{d}_t) \\ \le L_V\, L_d\, \mu_w + L_{V,d}(2\mu_w + L_w T_c). \end{aligned} \tag{43}$$

Proof of Lemma 2. Decompose

$$\begin{aligned} \Delta V &:= V_{t+1}(\mathbf{x}_{t+1}; \hat{d}_{t+1}) - V_{t+1}(\hat{\mathbf{x}}_{t+1}; \hat{d}_t) \\ &= \underbrace{V_{t+1}(\mathbf{x}_{t+1}; \hat{d}_{t+1}) - V_{t+1}(\mathbf{x}_{t+1}; \hat{d}_t)}_{=:\,(A)\ \text{estimate-update}} \\ &\quad + \underbrace{V_{t+1}(\mathbf{x}_{t+1}; \hat{d}_t) - V_{t+1}(\hat{\mathbf{x}}_{t+1}; \hat{d}_t)}_{=:\,(B)\ \text{state-mismatch}}. \end{aligned}$$

State-mismatch term (B). $\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1} = f_d(\mathbf{x}_t, \mathbf{u}_{t|t}, \mathbf{w}_t) - f_d(\mathbf{x}_t, \mathbf{u}_{t|t}, \hat{d}_t)$, so by Assumption 4 and $\|\mathbf{w}_t - \hat{d}_t\| \le \mu_w$ from (18), $\|\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1}\| \le L_d\,\mu_w$. Hence $|(B)| \le L_V\, L_d\, \mu_w$.

Estimate-update term (A). The triangle inequality on $\hat{d}_{t+1} - \hat{d}_t = (\hat{d}_{t+1} - \mathbf{w}_{t+1}) + (\mathbf{w}_{t+1} - \mathbf{w}_t) + (\mathbf{w}_t - \hat{d}_t)$ together with Assumption 1 gives $\|\hat{d}_{t+1} - \hat{d}_t\| \le 2\mu_w + L_w T_c$, where $L_w := L_{w_t} + L_{w_\nu}\phi$ bounds the total one-step variation of the disturbance along the trajectory. Hence $|(A)| \le L_{V,d}\,(2\mu_w + L_w T_c)$.

Summing the bounds on (A) and (B) yields (43). ■

Proof of Theorem 1. Combining Lemmas 1–2,

$$\begin{aligned} V_{t+1}(\mathbf{x}_{t+1}; \hat{d}_{t+1}) &= V_{t+1}(\hat{\mathbf{x}}_{t+1}; \hat{d}_t) + \Delta V \\ &\le \varepsilon\, V_t(\mathbf{x}_t; \hat{d}_t) + C, \end{aligned}$$

where $C := L_V\, L_d\, \mu_w + L_{V,d}(2\mu_w + L_w T_c)$. Writing $V_t := V_t(\mathbf{x}_t; \hat{d}_t)$ for brevity, induction in t gives

$$V_t \le \varepsilon^{t-1} V_1 + C \sum_{k=0}^{t-2} \varepsilon^k, \quad t \ge 1. \tag{44}$$

Since $\varepsilon \in (0,1)$, the finite geometric sum in (44) satisfies $\sum_{k=0}^{t-2} \varepsilon^k \le 1/(1-\varepsilon)$, so $V_t \le \varepsilon^{t-1} V_1 + C/(1-\varepsilon)$. Applying the value-function upper bound $V_1 \le \bar{\lambda}\,\sigma(\mathbf{x}_1 - \mathbf{r}_1)$ and the implied lower bound $\underline{\lambda}\,\sigma(\mathbf{x}_t - \mathbf{r}_t) \le V_t$ of Assumption 3 yields

$$\sigma(\mathbf{x}_t - \mathbf{r}_t) \le \tfrac{\bar{\lambda}}{\underline{\lambda}} \varepsilon^{t-1} \sigma(\mathbf{x}_1 - \mathbf{r}_1) + \tfrac{C}{\underline{\lambda}(1-\varepsilon)},$$

which is (21). Since the first term decays geometrically, $\mathbf{x}_t$ enters the residual set $\mathcal{B} = \{\mathbf{x} : \sigma(\mathbf{x} - \mathbf{r}) \le C/[\underline{\lambda}(1-\varepsilon)]\}$ in finite time and remains there, which establishes ultimate boundedness. ■

References

[1] C. Lee, D. Chung, J. Kim, and J. Kim, "Nonlinear model predictive control with obstacle avoidance constraints for autonomous navigation in a canal environment," *IEEE/ASME Transactions on Mechatronics*, vol. 29, no. 3, pp. 1985–1996, 2024.

[2] A. B. Martinsen, A. M. Lekkas, and S. Gros, "Reinforcement learning-based nmpc for tracking control of asvs: Theory and experiments," *Control Engineering Practice*, vol. 120, p. 105024, 2022.

[3] W. Wang, D. Fernández-Gutiérrez, R. Doornbusch, J. Jordan, T. Shan, P. Leoni, N. Hagemann, J. K. Schiphorst, F. Duarte, C. Ratti *et al.*, "Roboat iii: An autonomous surface vessel for urban transportation," *Journal of Field Robotics*, vol. 40, no. 8, pp. 1996–2009, 2023.

[4] L. Xu, B. Tian, C. Wang, J. Lu, D. Wang, Z. Li, and Q. Zong, "Fixed-time disturbance observer-based mpc robust trajectory tracking control of quadrotor," *IEEE/ASME Transactions on Mechatronics*, 2024.

[5] J. Kim and M. J. Kim, "Disturbance-aware model predictive control of underactuated robotics systems," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 11 985–11 992.

[6] J. Köhler, R. Soloperto, M. A. Müller, and F. Allgöwer, "A computationally efficient robust model predictive control framework for uncertain nonlinear systems," *IEEE Transactions on Automatic Control*, vol. 66, no. 2, pp. 794–801, 2020.

[7] J. Köhler, P. Kötting, R. Soloperto, F. Allgöwer, and M. A. Müller, "A robust adaptive model predictive control framework for nonlinear uncertain systems," *International Journal of Robust and Nonlinear Control*, vol. 31, no. 18, pp. 8725–8749, 2021.

[8] S. Singh, A. Majumdar, J.-J. Slotine, and M. Pavone, "Robust online motion planning via contraction theory and convex optimization," in *2017 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2017, pp. 5883–5890.

[9] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Transactions on Automatic Control*, vol. 62, no. 8, pp. 3861–3876, 2016.

[10] W. Xiao and C. Belta, "High-order control barrier functions," *IEEE Transactions on Automatic Control*, vol. 67, no. 7, pp. 3655–3662, 2022.

[11] M. H. Cohen and C. Belta, "High order robust adaptive control barrier functions and exponentially stabilizing adaptive control Lyapunov functions," in *2022 American Control Conference (ACC)*. IEEE, 2022, pp. 2233–2238.

[12] Q. Nguyen and K. Sreenath, "Robust safety-critical control for dynamic robotics," *IEEE Transactions on Automatic Control*, vol. 67, no. 3, pp. 1073–1088, 2022.

[13] E. I. Sarda and M. R. Dhanak, "Launch and recovery of an autonomous underwater vehicle from a station-keeping unmanned surface vehicle," *IEEE Journal of Oceanic Engineering*, vol. 44, no. 1, pp. 290–299, 2019.

[14] Y. Liao, C. Chen, T. Du, J. Sun, Y. Xin, Z. Zhai, B. Wang, Y. Li, and S. Pang, "Research on disturbance rejection motion control method of usv for uuv recovery," *Journal of Field Robotics*, vol. 40, no. 3, pp. 574–594, 2023.

[15] T. Xie, Y. Li, Y. Jiang, S. Pang, and X. Xu, "Three-dimensional mobile docking control method of an underactuated autonomous underwater vehicle," *Ocean Engineering*, vol. 265, p. 112634, 2022.

[16] S. Xu, J. Wu, C. Long, L. Wang, and S. Li, "Multi-stage path planning for unmanned surface vessels recovery," in *2023 American Control Conference (ACC)*. IEEE, 2023, pp. 4605–4610.

[17] K. Park, C. Lee, and J. Kim, "Robust trajectory tracking for surface vehicles via L1-augmented model predictive control," in *OCEANS 2025 Brest*. IEEE, 2025.

[18] A. Pavlov, H. Nordahl, and M. Breivik, "Mpc-based optimal path following for underactuated vessels," *IFAC Proceedings Volumes*, vol. 42, no. 18, pp. 340–345, 2009.

[19] Z. Wu, S. Cheng, P. Zhao, A. Gahlawat, K. A. Ackerman, A. Lakshmanan, C. Yang, J. Yu, and N. Hovakimyan, "$\mathcal{L}_1$quad: $\mathcal{L}_1$ adaptive augmentation of geometric control for agile quadrotors with performance guarantees," *IEEE Transactions on Control Systems Technology*, vol. 33, no. 2, pp. 597–612, 2025.

[20] D. Hanover, P. Foehn, S. Sun, E. Kaufmann, and D. Scaramuzza, "Performance, precision, and payloads: Adaptive nonlinear MPC for quadrotors," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 690–697, 2022.

[21] Y. Gao, A. Gray, H. E. Tseng, and F. Borrelli, "A tube-based robust nonlinear predictive control approach to semiautonomous ground vehicles," *Vehicle System Dynamics*, vol. 52, no. 6, pp. 802–823, 2014.

[22] B. A. H. Vicente, P. A. Trodden, and S. R. Anderson, "Fast tube model predictive control for driverless cars using linear data-driven models," *IEEE Transactions on Control Systems Technology*, vol. 31, no. 3, pp. 1395–1410, 2022.

[23] R. Tao, P. Zhao, I. Kolmanovsky, and N. Hovakimyan, "Robust adaptive mpc using uncertainty compensation," in *2024 American Control Conference (ACC)*. IEEE, 2024, pp. 1873–1878.

[24] V. N. Sankaranarayanan, A. Saradagi, S. Satpute, and G. Nikolakopoulos, "Time-varying control barrier function for safe and precise landing of a UAV on a moving target," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024.

[25] ——, "Collision-free landing of multiple UAVs on moving ground vehicles using time-varying control barrier functions," in *2024 American Control Conference (ACC)*. IEEE, 2024.

[26] A. Saradagi, V. N. Sankaranarayanan, A. Banerjee, S. Satpute, and G. Nikolakopoulos, "Switched control barrier functions-based safe docking control strategy for a planar floating platform," *Control Engineering Practice*, vol. 158, p. 106274, 2025.

[27] D. Zeng, Y. Jiang, Y. Wang, H. Zhang, and Y. Feng, "Robust adaptive control barrier functions for input-affine systems: Application to uncertain manipulator safety constraints," *IEEE Control Systems Letters*, vol. 8, 2024.

[28] D. R. Agrawal and D. Panagou, "Safe control synthesis via input constrained control barrier functions," in *2021 60th IEEE Conference on Decision and Control (CDC)*. IEEE, 2021, pp. 6113–6118.

[29] S. Liu, C. Liu, and J. Dolan, "Safe control under input limits with neural control barrier functions," in *Conference on Robot Learning*. PMLR, 2023, pp. 1970–1980.

[30] J. Zeng, B. Zhang, and K. Sreenath, "Safety-critical model predictive control with discrete-time control barrier function," in *2021 American Control Conference (ACC)*. IEEE, 2021, pp. 3882–3889.

[31] C. Lee, K. Park, and J. Kim, "Turning circle-based control barrier function for efficient collision avoidance of nonholonomic vehicles," *arXiv preprint arXiv:2503.20280*, 2025.

[32] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. Chichester, UK: John Wiley & Sons, 2011.

[33] R. Verschueren, G. Frison, D. Kouzoupis, J. Frey, N. van Duijkeren, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, "acados—a modular open-source framework for fast embedded optimal control," *Mathematical Programming Computation*, vol. 14, no. 1, pp. 147–183, 2022.

[34] G. Frison and M. Diehl, "Hpipm: a high-performance quadratic programming framework for model predictive control," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 6563–6569, 2020.

[35] S. Liu, J. Zeng, K. Sreenath, and C. A. Belta, "Iterative convex optimization for model predictive control with discrete-time high-order control barrier functions," in *2023 American Control Conference (ACC)*. IEEE, 2023, pp. 3368–3375.

[36] Z. Shayan, M. Izadi, V. Scognamiglio, S. D'Angelo, S. Singoji, V. Lippiello, and R. Faieghi, "Exponential control barrier function and model predictive control for jerk-level reactive motion planning of quadrotors," *Control Engineering Practice*, vol. 164, p. 106489, 2025.

[37] G. Grimm, M. J. Messina, S. E. Tuna, and A. R. Teel, "Model predictive control: for want of a local control Lyapunov function, all is not lost," *IEEE Transactions on Automatic Control*, vol. 50, no. 5, pp. 546–558, 2005.

[38] D. Q. Mayne, E. C. Kerrigan, E. Van Wyk, and P. Falugi, "Tube-based robust nonlinear model predictive control," *International journal of robust and nonlinear control*, vol. 21, no. 11, pp. 1341–1353, 2011.

[39] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. Madison, WI: Nob Hill Publishing, 2017.

[40] T. I. Fossen, "An adaptive line-of-sight (ALOS) guidance law for path following of aircraft and marine craft," *IEEE Transactions on Control Systems Technology*, vol. 31, no. 6, pp. 2887–2894, 2023.